WILL

Relational Orbital Mechanics (R.O.M.)

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Abstract

This supplementary document to the first part of the WILL trilogy presents **Relational Orbital Mechanics (R.O.M.)** – a closed algebraic system that classifies allowed states of relational configuration without invoking a background spacetime, a metric tensor, or a mass parameter and gravitational constant.

This closed algebraic system recovers the leading contributions to perihelion precession, gravitational and transverse Doppler shifts, light deflection and frame-dragging through elementary closed-form expressions.

High-precision validation shows that all 66 multi-form identities are internally consistent to 50 significant digits and that the leading terms match the corresponding post-Newtonian expansions of general relativity. Higher-order differences remain below the sensitivity of current observations, including those of the S2 star.

Complete phase-resolved orbits and system scales are recovered directly from spectroscopic and chronometric observables, with G and M entering only as unit-conversion factors. Key features of Kerr geometry follow algebraically from the same closure relation. Gravitational waves and the radiative two-body problem remain to be formulated within the relational ontology.

From the dimensionless projections β (kinematic) and κ (potential) on the relational carriers S^1 and S^2, and from the closure theorem $\kappa^2 = 2\beta^2$ (forced by degree-of-freedom indifference), all major observed orbital phenomena are derived algebraically as an inevitable consequence of Relational Geometry:

- Kepler's Third Law, the vis-viva relation, conservation of angular momentum, and the derivation of eccentricity emerge as strict Pythagorean invariants.
- Orbital precession is recovered as the accumulation of the relational phase divergence τ_Y^2, normalized by the orbital shape factor $1 - e^2$, yielding the GR value for Mercury without differential equations.
- The Schwarzschild radius R_s and all orbital scales are expressed directly in terms of spectroscopic and chronometric observables ($z_\odot$, $\theta_\odot$, T, e), bypassing mass M and Newton's constant G.
- Gravitational deflection of massive particles and light follows from the energy symmetry law with no weak-field expansion.

R.O.M. demonstrates that the operational content of gravitational dynamics is fully captured by the algebra of relational projections. The construct serves as a rigorous, empirical realization of the foundational methodological principles established in WILL Part I Relational Geometry. As Open Research, all data and calculation tools are publicly available at willrg.com and https://github.com/AntonRize/WILL.

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Contents

1 Quick Recap

In the following sections we will demonstrate that RG's relational projections (κ, β) are ontologically and operationally independent from mass M and gravitational constant G. We will show that R_s as a horizon scale is derived directly from the phase relations of light, and how all observational phenomena of orbital systems are generated as unavoidable consequences of Relational Geometry.

But first, let us remind ourselves of the key derivations and the logical chain established prior:

Derivation Chain Map

Methodological constraints:

↓(*Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency*)↓

Relational Closure + Causal Continuity + Isotropy Theorems

↓ *(apply to existing physics)* ↓

Standard Primitives: $\underbrace{\text{background manifold} + \text{metric}}_{\text{spacetime structure (not derived)}} \quad + \quad \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$

↓ *(Ontological reduction by Relational Origin Principle — no background allowed)* ↓

One Primitive: ***WILL*** $\equiv$ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ $\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$

↓ *(by Relational Closure + Causal Continuity + Isotropy: derive relational energy carriers)* ↓

S^1 **(1-DOF kinematic) +** S^2 **(2-DOF potential)**

↓ *(by Duality of Relation Lemma)* ↓

Relational Conservation Theorem: $\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$

$\boxed{\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}} \quad \underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic } S^1}}$ **Relational Projections** $\boxed{\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential } S^2} \quad \underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}}$

↓ *(by DOF-Indifference Lemma: equal quadratic weight to each independent DOF)* ↓

Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \Longrightarrow \boxed{\kappa^2 = 2\beta^2}$

↓ *(by Relational Conservation Theorem derive metric intervals as scaffolding)* ↓

Minkowski: $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$

Schwarzschild: $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 (1 - (R_s/r))\, dt^2}_{\text{mathematical scaffolding}}$

↓ *(by Relational Closure and Causal Continuity Theorems: any change must be balanced)* ↓

Energy Symmetry Law: $\boxed{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}$

↓ $\Delta E_{A\to B} = E_0(\kappa_{X,B}/\beta_{Y,B} - \kappa_{X,A}/\beta_{Y,A})$; $\Delta E_{B\to A} = E_0(\kappa_{X,A}/\beta_{Y,A} - \kappa_{X,B}/\beta_{Y,B})$ ↓

Speed of light: $\beta \to 1 \;\to\; \beta_Y \to 0 \;\to\; E = E_0 \cdot \kappa_X / 0 \to \infty \;\to$ **requires** ∞ **energy**

Event Horizon: $\kappa \to 1 \;\to\; \kappa_X \to 0 \;\to\; E = E_0 \cdot 0/\beta_Y \to 0 \;\to$ **requires** 0 **energy**

↓ *(by Unified Relational Scaling Lemma)* ↓

Equivalence Principle $\boxed{m_g \equiv m_i \equiv m = E_0/c^2}$

↓ *(**Lagrangian and Hamiltonian** derived as collapsed single-point approximations)* ↓

$\underbrace{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}_{\text{exact symmetry law}} \approx \underbrace{\tfrac{1}{2}(\kappa_A^2 - \kappa_B^2) + \tfrac{1}{2}(\beta_B^2 - \beta_A^2)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$

↓ *(apply RG to orbital mechanics)* ↓

Relational Orbital Mechanics $\boxed{\textbf{R.O.M.}}$**: a closed algebraic system of equations**

↓*(by Factorization Theorem: every quantity X splits into unitless core $\hat{X}$ and scale label S_X)*↓

Unitless Core: $X = \underbrace{\hat{X}(,e)}_{\text{unitless core}} \underbrace{S_X(R_s, c, G)}_{\text{units-scale}}$ **G and M are unit-conversion factors.**

↓ *(by Inverse Square, Closure and Causal Continuity Theorems)* ↓

No Singularities: $\beta_{\max}^2 = 1 \;\to\; \kappa_{\max}^2 = 2 \;\to\; r_{\min} = R_s/\kappa_{\max}^2 \;\to\; r_{\min} = R_s/2$

The point $r = 0$ **is absent from the admissible relational range**

↓ *(apply 2D to 3D translation interfaces to derive Relational Field Equation:)* ↓

$\boxed{\kappa^2 = R_s/r = \rho_{\text{field}}/\rho_{\max}} \Longrightarrow \boxed{\text{Spacetime Geometry}(R_s/r)} \equiv \boxed{\text{Energy Density}(\rho/\rho_{\max})}$

↓ **Theoretical Ouroboros** ↓

$\boxed{\text{SPACETIME} \equiv \text{ENERGY}} \Leftrightarrow \boxed{\text{SPACETIME GEOMETRY} \equiv \text{ENERGY DENSITY}}$

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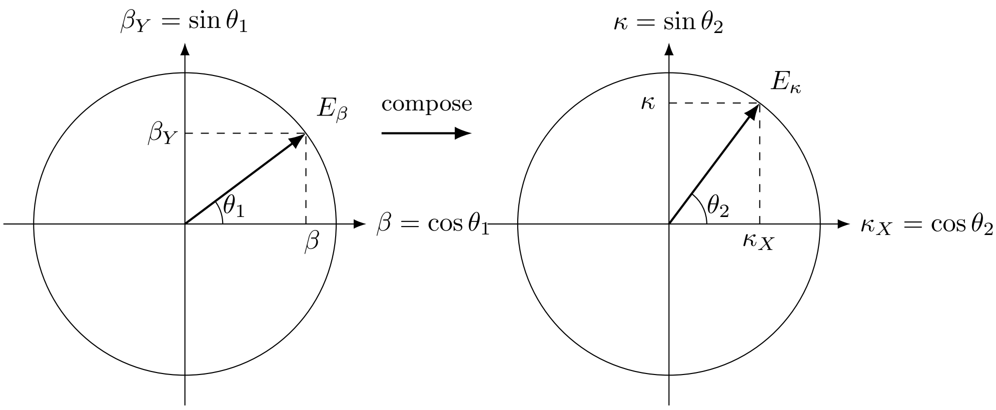

$E_\beta = E_0/\beta_Y$
$p_\beta = E_0\,\beta/\beta_Y = E_0\cot(\theta_1)$

$E_\kappa = E_0/\kappa_X$
$p_\kappa = E_0\,\kappa/\kappa_X = E_0\tan(\theta_2)$

2 Closed Algebraic System of R.O.M. Equations

This section presents the closed algebraic system of Relational Orbital Mechanics (R.O.M.). Derivations of all quantities are traceable back to Methodological Principles and two relational primitives (β on S^1, κ on S^2) linked by the Closure Theorem $\kappa^2 = 2\beta^2$. The first block gives global invariants fixed for an entire orbit; subsequent blocks give phase-dependent quantities. Everything is expressed in direct observables (observed ratios, spectroscopic, chronometric and astrometric readings) so that no background metric is required.

Remark 2.1. *Instead of postulating that a body moves under forces through a preexisting container-like spacetime, R.O.M. derives the unique, algebraically allowed relational states of a system. With no need to postulate additional degrees of freedom, the system has no use for variational or least-action principles. It is not a set of equations of motion but an algebraically closed system of allowed states within a relational system.*

Relational Unitless Core Inputs and Degrees of Freedom

Remark 2.2. *The configurations below are operational examples rather than an exhaustive taxonomy; algebraic closure ensures that* ***any set of parameters*** *spanning the system's* ***degrees of freedom*** *serves as a* ***complete input basis****.*

The unitless relational core possesses three degrees of freedom: **scale**, **shape**, and **phase**, which are closed by any of the following input sets:

- **Global Invariant Basis (2 inputs):**
  (β, e) or (κ, e). Directly determines the system invariant baseline and orbital geometry.
- **Arbitrary Local State Basis (3 inputs):**
  - Decomposed Kinematics: $(\kappa_o, \beta_T, \beta_R)$
  - Scalar Kinematics with Phase: (κ_o, β_o, O)

  Closes the state at an arbitrary orbital position without symmetry constraints.
- **Boundary-Constrained Basis (2 inputs):**
  - Perihelion: (κ_p, β_p), where $\beta_R = 0$ and $O = 0$.
  - Aphelion: (κ_a, β_a), where $\beta_R = 0$ and $O = \pi$.
  - Balance Point: (κ, B_a) or (κ, β_R), where $\eta_o = 1$ and $\kappa_o^2 = 2\beta_o^2$.

  The geometric boundary condition eliminates one degree of freedom, closing the system from two parameters.

Historical Units Scale

Every unit-full quantity X of the closed R.O.M. system factorizes as

$$\boxed{X \;=\; \hat{X}(\beta,\,\kappa,\,e,\,O,\,i,\omega_i,\,\kappa_R)\;\cdot\;S_X(R_s;\,c,\,G)}$$

where $\hat{X}$ is a closed-form pure number determined entirely by dimensionless relational inputs, and S_X is a monomial scale factor depending only on the quantity class:

Quantity class	S_X	**Quantity class**	S_X
projections, shifts, angles, phases	1	frequency	c/R_s
length	R_s	specific angular momentum	$R_s\,c$
time	R_s/c	mass	$R_s \cdot c^2/G$
velocity	c	density	c^2/GR_s^2

No dimensional constant enters any $\hat{X}$. All dynamics are solved within the relational unitless core. All physics takes place within the relational core. Units and constants serve only as external labels for human measurement systems.

Colab Notebook R.O.M. Tests

ROM_FULL_TEST.ipynb - In this notebook we perform 3 wide tests of Relational Orbital Mechanics:

- Algebraically test the closed system (internal consistency of every multi-form identity). All 66 expressions (50-digit, random c,G,M,a,e,O).
- R.O.M. vs GR: We compute the exact Schwarzschild geodesic results by high-precision numerical integration (no approximation), then extract their Taylor coefficients in the SAME small parameter and compare term-by-term to R.O.M..
- ROM reconstructs a, R_s, r_p, r_a and the whole phase-resolved orbit, and recovers GM (a length^3/ time^2 combination, no G) and finally M (kg, G used ONLY here as a units conversion factor).

Test it yourself

DESMOS: https://www.desmos.com/calculator/gr7qykzicf interactive user-friendly R.O.M. solver.

$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}}$ z_κ = gravitational redshift

$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}}$ z_β = transverse Doppler shift

Observational Inputs

$Z_{sys} = (1+z_\kappa)(1+z_\beta) = \frac{1}{\tau}$ (product of gravitational red shift and transverse Doppler shift)
$\tau = \kappa_X \cdot \beta_Y = \frac{1}{Z_{sys}}$ (product of projectional phase factors on S^1 and S^2 carriers)
$z_\kappa = \frac{1}{\kappa_X} - 1$ (gravitational redshift)
$z_\beta = \frac{1}{\beta_Y} - 1$ (transverse Doppler shift)
$z_{\kappa o} = \frac{1}{\kappa_{Xo}} - 1$ (redshift at phase O)
$z_{\beta o} = \frac{1}{\beta_{Yo}} - 1$ (transverse Doppler shift at phase O)

Global Unit-Free System Parameters (Fixed for the Orbit)

$\kappa = \sin(\theta_2) = \beta\sqrt{2} = \sqrt{1-(1+z_\kappa)^{-2}} = \sqrt{\frac{R_s}{a}} = \sqrt{\frac{\rho}{\rho_{max}}} = \sqrt{\kappa_p^2(1-e)} = \sqrt{2(\kappa_o^2-\beta_o^2)} = \sqrt{\frac{2\cdot a\cdot g}{c^2}} = \sqrt{\frac{1}{2}(3-\sqrt{1+8\tau_o(B_a)^2})} = \kappa_{sur}\sqrt{\sin(\theta_{sur})} = \kappa_{sur}(\frac{\tau_D}{T})^{\frac{1}{3}} = \sqrt{\kappa_o^2\cdot\eta_o}$ (potential projection at semi-major axis)
$\kappa_X = \cos(\theta_2) = \sqrt{1-\kappa^2} = \frac{1}{(1+z_\kappa)}$ (potential phase factor)
$\kappa_{sur} = \sqrt{\frac{\kappa^2}{\sin(\theta_{sur})}} = \sqrt{\frac{2\pi}{\beta\sin(\theta_{sur})}\cdot\frac{R_s}{Tc}}$ (potential projection at the surface of central body)
$\beta = \cos(\theta_1) = \sqrt{1-(1+z_\beta)^{-2}} = \frac{\kappa}{\sqrt{2}} = \beta_p e_X^{-\frac{1}{2}} = \frac{2\pi a}{Tc} = (\frac{\pi R_s}{Tc})^{\frac{1}{3}} = \sqrt{(\kappa_o^2-\beta_o^2)} = \sqrt{\frac{\kappa_p^2}{2}(1-e)} = \beta_o\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\cdot e\cdot\cos(O)}} = \sqrt{\beta_{sys}^2\cdot\sin(\theta_{sys})}$ (global kinetic projection (invariant transverse baseline defining the system's circular equilibrium state and semi-major axis))
$\beta_Y = \sin(\theta_1) = \sqrt{1-\beta^2} = \frac{1}{(1+z_\beta)}$ (kinetic phase factor)
$\beta_{int} = \frac{\beta}{e_Y}$ (interior kinetic projection)
$\tau = \kappa_X\beta_Y = \sqrt{(1-\kappa^2)\cdot(1-\beta^2)}$ (relational spacetime factor)
$\tau_Y = \sqrt{1-\tau^2} = \sqrt{3\beta^2-2\beta^4} = \sqrt{\kappa^2+\beta^2-\kappa^2\beta^2}$ (relational spacetime phase factor)
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{\frac{3}{2}}\ \kappa = \sqrt{3}\ \beta$ (total relational shift (magnitude of state difference))
$\beta^2 = (\kappa_o^2-\beta_o^2) = \frac{\kappa_p^2}{2}(1-e) = (\kappa_o^2-\frac{\kappa_o^2}{2\delta_o}) = \frac{R_s}{2a} = (\frac{R_s}{T\cdot c}\pi)^{\frac{2}{3}}$ (energy invariant - binding energy)
$\Delta\phi = \frac{2\pi\cdot\tau_Y^2}{1-e^2} = \frac{2\pi(3\beta^2-2\beta^4)}{1-e^2} = \frac{3\pi R_s}{a(1-e^2)} - \frac{\pi R_s^2}{a^2(1-e^2)}$ (precession of perihelion per orbit)

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$\Omega_\phi = \frac{\tau_Y^2}{1-e^2} = \frac{3\beta^2-2\beta^4}{1-e^2}$ (precession of perihelion per phase radian)
$\Omega = 1 - \Omega_\phi = \frac{O}{o}$ (constant ratio between idealised phase o and true precessing phase O)
$\Delta\varphi = 2\arcsin(\frac{\kappa_p^2(1+\beta_p^2)}{2\beta_p^2-\kappa_p^2(1+\beta_p^2)})$ (gravitational deflection)
$\Delta\gamma = 2\arcsin(\frac{\kappa_p^2}{\kappa_{Xp}^2})$ (light deflection)

Global Unit-Full System Parameters

$R_s = \frac{\beta^3}{\pi} \cdot Tc = \frac{\beta_{sur}^3}{\pi} \cdot \tau_D c = \kappa_o^2 \cdot r_o = \frac{\kappa^2}{\zeta}\beta \cdot \Delta_{to} c = \frac{8\pi^2 a^3}{T^2 c^2} = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2) = \frac{a}{2}(3 - \sqrt{1 + 8\tau_o(B_a)^2}) = \frac{r_o}{2(2a-r_o)}(4a - r_o - ((4a-r_o)^2 - 8a(2a-r_o)\ (1-\tau_o^2))^{\frac{1}{2}}) = \frac{2GM}{c^2}$ (Schwarzschild radius - system scale)
$E_0 = \frac{1}{2} \cdot R_s \frac{c^4}{G}$ (rest energy)
$E = \frac{\kappa_X}{2\beta_Y} \cdot R_s \frac{c^4}{G}$ (total relational energy scaled by kinetic and potential projections)
$E_\beta = \frac{1}{2\beta_Y} \cdot R_s \frac{c^4}{G}$ (energy on S^1 relational carrier)
$E_\kappa = \frac{1}{2\kappa_X} \cdot R_s \frac{c^4}{G}$ (energy on S^2 relational carrier)
$p_\beta = \frac{\beta}{2\beta_Y} \cdot R_s \frac{c^3}{G}$ (kinetic momentum)
$p_\kappa = \frac{\kappa}{2\kappa_X} \cdot R_s \frac{c^3}{G}$ (potential momentum)
$a = \frac{1}{\kappa^2} \cdot R_s = \frac{T\beta c}{2\pi} = \frac{\beta_o c}{\omega} \cdot \frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e\ cos(O)}} = \frac{\Delta_{to}(O)}{\zeta}\beta c = \frac{Tc\beta^3}{2\pi(\frac{K_i}{\sin(i)})^2(1-e^2)} = \frac{\sqrt{1-e^2}}{2\pi\sin(i)} K_i T c = \frac{R_{sys}}{\sin(\theta_{sys})}$ (semi-major axis)
$h = \frac{e_Y}{2\beta} \cdot R_s c = \frac{\kappa^2}{\kappa_o^2} \cdot \beta_T \cdot a \cdot c = r_o \cdot \beta_T \cdot c = r_o^2 \cdot \omega_o = a \cdot \beta c \cdot e_Y$ (invariant angular momentum)
$\omega = 2\beta^3 \cdot \frac{c}{R_s} = \frac{\beta c}{a}$ (angular frequency)
$T = \frac{\pi}{\beta^3} \cdot \frac{R_s}{c} = \frac{2\pi\sqrt{2}}{\kappa^3} \cdot \frac{R_s}{c} = \frac{2\pi}{\beta} \cdot \frac{a}{c} = \frac{2\pi}{\omega} = \frac{2\pi\Delta_{to}}{\zeta}$ (orbital period)
$\tau_D = \frac{\pi\sin(\theta_{sur})^{\frac{3}{2}}}{\beta^3} \cdot \frac{R_s}{c} = T\sin(\theta_{sur})^{\frac{3}{2}} = \frac{2\pi R_{sur}}{\beta_{sur} c} = \sqrt{\frac{\pi}{G \cdot \rho_{sur}(R_{sur})}}$ (time-density invariant where R_{sur} and β_{sur} determined by the distance from center to the surface of central body and θ_{sur} = angular radius)
$g = \frac{\kappa^4}{2} \cdot \frac{c^2}{R_s} = \frac{\kappa^2 c^2}{2a} = \frac{GM}{a^2}$ (gravitational acceleration)
$M = \frac{1}{2} \cdot \frac{R_s c^2}{G} = \frac{\beta^3}{2\pi} \cdot T\frac{c^3}{G} = \beta^2 \cdot a\frac{c^2}{G} = 4\pi\rho a^3$ (mass parameter)
$\rho = \frac{\kappa^6}{8\pi} \cdot \frac{c^2}{GR_s^2} = \frac{1}{8\pi}(\frac{z_k(z_k+2)}{(1+z_k)^2})^3 \cdot \frac{c^2}{G \cdot R_s^2} = \frac{\kappa^2 \cdot c^2}{8 \cdot \pi \cdot G \cdot a^2}$ (relational density)
$\rho_{sur} = \frac{\kappa_{sur}^6}{8\pi} \cdot \frac{c^2}{GR_s^2} = \frac{\pi}{G \cdot \tau_D^2} = \frac{\kappa_{sur}^2 \cdot c^2}{8 \cdot \pi \cdot G \cdot R_{sur}^2}$ (relational density of the central body within (R_{sur}) its physical radius)
$\rho_{max} = \frac{1}{8\pi} \cdot \frac{c^2}{GR_s^2}$ (maximal density at the horizon (for non-rotating systems), $\kappa^2 = 1 \equiv r = R_s$; density reaches its natural bound)
$R_{sur} = \frac{\sin(\theta_{sur})}{\kappa^2} \cdot R_s = a \cdot \sin(\theta_{sur}) = \sqrt[3]{\frac{\pi a^3}{GT^2\rho(R_{sur})}}$ (system's central body's physical radius)
$t = \frac{1}{\kappa^2} \cdot \frac{R_s}{c} = \frac{a}{c}$(temporal radius of the system)

Eccentricity Relations

$e = \frac{2\beta_p^2}{\kappa_p^2} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2} = \frac{1}{\delta_p} - 1 = \frac{r_a - r_p}{r_p + r_a} = 1 - \eta_o(0) = \eta_o(\pi) - 1 = \sqrt{1 - \frac{\beta_T^2 \cdot \eta_o^2}{\beta^2}} = \frac{1}{2} \cdot (-\eta_o\cos(O) + \sqrt{\eta_o^2\cos^2(O) - 4(\eta_o - 1)})$ (eccentricity derived from closure theorem)
$e_Y = \sqrt{1-e^2}$ (eccentricity's orthogonal complement)
$e_X = \frac{1+e}{1-e} = \frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (shape factor)

Time-phase

$\zeta = \frac{2\pi\Delta_t}{T} = \frac{\Delta_t c}{R_s} 2\beta^3 = (1-e^2)^{\frac{3}{2}} \cdot (\int_0^O (1 + e \cdot \cos(\theta))^{-2} d\theta) = \frac{1}{\sqrt{1-e^2}} \int_0^O (\frac{t_o(\theta)}{t})^2 d\theta$ (Temporal Phase interval (mean anomaly analog))
$\zeta(<\pi) = 2\arctan(\frac{\sin(\frac{O}{2})}{e_X^{\frac{1}{2}}\cos(\frac{O}{2})}) - \frac{e \cdot e_Y\sin(O)}{1+e\cos(O)}$ (Temporal Phase interval ascending orbit from 0 to π)
$\zeta(>\pi) = 2\arctan(\frac{\sin(\frac{O}{2})}{e_X^{\frac{1}{2}}\cos(\frac{O}{2})}) - \frac{e \cdot e_Y\sin(O)}{1+e\cos(O)} + 2\pi$ (Temporal Phase interval descending orbit from π to 2π)
$\zeta_B = \left(1 + \frac{\beta^2}{2} - \kappa^2\right) \cdot 2\arctan\left(\frac{\sin(\frac{O}{2})}{e_X^{\frac{1}{2}}\cos(\frac{O}{2})}\right) - \left(1 + \frac{\beta^2}{2}\right) \cdot \frac{e \cdot e_Y\sin(O)}{1+e\cos(O)}$ (Coordinate Temporal Phase interval ascending orbit from 0 to π)

$\omega_o = 2\beta^3\frac{(1+e\cos(O))^2}{(1-e^2)^{3/2}} \cdot \frac{c}{R_s} = \frac{\beta \cdot c}{a} \cdot \frac{(1+e \cdot cos(O))^2}{(1-e^2)^{\frac{3}{2}}}$ (angular frequency at phase O)
$\Delta_{to} = \frac{\zeta}{2\beta^3} \cdot \frac{R_s}{c} = \int_0^O \frac{1}{\omega_o(\theta)} d\theta = \frac{a}{\beta c}\zeta = \frac{T}{2\pi}\zeta$ (time duration of given phase interval)

Perihelion Relations

$\kappa_p = \kappa\sqrt{\frac{1}{1-e}} = \sqrt{\frac{2\beta^2}{1-e}} = Q_p\sqrt{\frac{2}{3+e}}$ (potential projection at perihelion)
$\kappa_{Xp} = \sqrt{1-\kappa_p^2}$ (potential phase projection at perihelion)
$\beta_p = \frac{V_p}{c} = \beta \ \cdot e_X^{1/2} = \beta_a \cdot e_X = \frac{r_p\omega_o(0)}{c} = \frac{\kappa_p}{\sqrt{2}}\sqrt{1+e}$ (kinetic projection at perihelion)
$\delta_p = \frac{\kappa_p^2}{2\beta_p^2} = \frac{1}{1+e}$ (closure factor at perihelion)
$Q_p = \sqrt{\kappa_p^2 + \beta_p^2}$ (relational shift at perihelion)

$r_p = \frac{1}{\kappa_p^2} \cdot R_s = a(1-e)$ (radius at perihelion)

Aphelion Relations

$\beta_a = \beta \cdot e_X^{-1/2} = \beta_p \cdot e_X^{-1}$ (kinetic projection at aphelion)
$\kappa_a = \sqrt{\beta^2 + \beta_a^2}$ (potential projection at aphelion)
$\delta_a = \frac{1}{1-e} = \frac{\kappa_a^2}{2\beta_a^2}$ (closure factor at aphelion)
$Q_a = \sqrt{\kappa_a^2 + \beta_a^2}$ (relational shift at aphelion)

$r_a = \frac{1}{\kappa_a^2} \cdot R_s = a(1+e)$ (radius at aphelion)

Phase Variables (Depend on phases o and O)

$o = \frac{O}{\Omega}$ (idealised orbital phase in radians, represent phase o progression without orbital precession)
$O = \Omega \cdot o = o - \Omega_\phi o$ (true orbital phase with precession)
$\eta_o = \frac{\kappa^2}{\kappa_o^2} = \frac{1-e^2}{1+e\cdot\cos(O)} = \frac{r_o}{a} = \frac{R_s}{\kappa_o^2 a} = 2 - \frac{2\beta_o^2}{\kappa_o^2}$ (normalised true radial distance at phase O)
$\kappa_o = \kappa \cdot \sqrt{\frac{1+e\cdot\cos(O)}{1-e^2}} = \sqrt{\beta_R^2 + \beta_T^2 + \beta^2} = \sqrt{\frac{R_s}{r_o}} = \sqrt{1-(1+z_{\kappa o})^{-2}} = \kappa_p\sqrt{\frac{1+e\cos(O)}{1+e}} = \kappa \ \cdot \eta_o^{-\frac{1}{2}} = \sqrt{\beta^2+\beta_o^2} = \sqrt{\kappa_{sur}^2 \cdot \sin(\theta_{sur,o})}$ (local amplitude potential projection at the phase O)
$\kappa_{Xo} = \sqrt{1-\kappa_o^2} = (z_{\kappa o}+1)^{-1}$ (potential phase projection at phase O)
$\beta_o = \sqrt{\beta_R^2 + \beta_T^2} = \beta \cdot \frac{\sqrt{1+e^2+2\cdot e\cdot\cos(O)}}{\sqrt{1-e^2}} = \sqrt{\kappa_o^2 - \beta^2} = \frac{\kappa_o}{\sqrt{2\delta_o}} = \sqrt{\frac{\kappa_o^2}{2}\frac{1+e^2+2\cdot e\cdot\cos(O)}{1+e\cdot\cos(O)}} = \sqrt{\frac{R_s}{r_o} - \frac{R_s}{2a}} = \sqrt{1-(1+z_{\beta o})^{-2}}$ (Instantaneous translational kinetic projection at phase O)
$\beta_R = \beta \ \frac{e\sin(O)}{\sqrt{1-e^2}} = \sqrt{\beta_o^2 - \beta_T^2}$ (Instantaneous radial kinetic projection at phase O)
$\beta_T = \frac{r_o\omega_o}{c} = \beta\frac{1+e\cos(O)}{\sqrt{1-e^2}} = \kappa_o^2\frac{\sqrt{1-e^2}}{2\beta} = \frac{R_s\sqrt{1-e^2}}{r_o 2\beta}$ (Instantaneous transverse kinetic projection at phase O)
$\beta_{Yo} = \sqrt{1-\beta_o^2} = (z_{\beta o}+1)^{-1}$ (kinetic phase projection at phase O)
$\delta_o = \frac{1+e\cos(O)}{1+e^2+2e\cos(O)} = \frac{\kappa_o^2}{2\beta_o^2}$ (local closure factor at phase O)
$Q_o = \sqrt{\kappa_o^2 + \beta_o^2}$ (local relational shift at phase O)
$\tau_o = \kappa_{Xo} \cdot \beta_{Yo} = Z_{sys,o}^{-1}$ (spacetime factor at phase O)
$\tau_{Yo} = \sqrt{1-\tau_o^2}$ (complement spacetime factor at phase O)

$r = r_o = \frac{1}{\kappa_o^2} \cdot R_s = a\frac{1-e^2}{1+e\cos(O)} = \frac{T}{2\pi} \cdot \beta c \cdot \frac{1-e^2}{1+e\cdot\cos(O)} = a \cdot \eta_o$ (radial distance at phase O)
$t_o = \frac{1}{\kappa_o^2} \cdot \frac{R_s}{c} = \frac{r_o}{c}$ (light time scale at phase O)

Orbital Phase Invariants

Holographic Decryption Invariant

$$Z_{raw}(-\omega_i)\tau(-\omega_i)(1 - e\cos\omega_i) + Z_{raw}(\pi-\omega_i)\tau(\pi-\omega_i)(1+e\cos\omega_i) = 2 \quad (1)$$

$\Delta E_{A\to B} + \Delta E_{B\to A} = 0$
$\beta^2 = (\kappa_o^2 - \beta_o^2)$
$\beta^2 - \beta_o^2 = \kappa^2 - \kappa_o^2$
$\frac{\beta_T}{\kappa_o^2} = \frac{e_Y}{2\beta}$
$\beta_T^2\eta_o^2 = \beta^2(1-e^2)$
$B_a - \zeta(B_a) = \zeta(B_d) - B_d$

$\frac{Z_{raw}(-\omega_i)}{Z_{sys}(-\omega_i)} = 1 + K_i\left(1 + e \cdot \cos\left(\omega_i\right)\right)$
$\frac{\kappa_o^2}{\kappa^2} = \frac{a}{r_o}$
$\beta \cdot \eta_o = \frac{2\pi \cdot t_o}{T}$
$\frac{\kappa_o^2}{\kappa^2} e_Y^2 = 1 + e\cos(O)$

Relational Geometry (WILL)

$\theta_1 = \arccos(\beta)$ (distribution angle on S^1, non-physical)
$\theta_2 = \arcsin(\kappa)$ (distribution angle on S^2, non-physical)
$\kappa^2 = 2\beta^2$ (Closure Theorem)
$\Delta E_{A\to B} + \Delta E_{B\to A} = 0 \quad \Delta E_{A\to B} = E_0(\frac{\kappa_{X,B}}{\beta_{Y,B}} - \frac{\kappa_{X,A}}{\beta_{Y,A}}); \quad \Delta E_{B\to A} = E_0(\frac{\kappa_{X,A}}{\beta_{Y,A}} - \frac{\kappa_{X,B}}{\beta_{Y,B}})$ (**Energy Symmetry Law** is the universal relational energy ledger, the origin of causality)
$\Delta_Q = Q_o^2 - Q^2$ (phase relational shift amplitude at phase O)
$B_a = \arccos(1 - \delta_p^{-1}) = \arccos(-e) = \arccos(1 - \frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2} - 1)$ (ascending orbital balance point where $a = r$ and $\kappa_o^2 = 2\beta_o^2$ are true)
$B_d = 2\pi - \arccos(-e)$ (descending orbital balance point where $a = r$ and $\kappa_o^2 = 2\beta_o^2$ are true)

Structural-Dynamical Equivalence

$\frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{1+e}{1-e} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (remarkable equivalence of structural (κ on S^2 carrier) and dynamical (β on S^1 carrier) symmetry suggesting once again that $SPACETIME \equiv ENERGY$)

Observer dependent

$\theta_{sur}(r) = \arcsin(\frac{R_{sur}}{r}) = \sqrt[3]{\frac{\pi}{\rho(R_{sur})GT^2}}$ (angular radius of central body viewed from distance r)
$Z_{raw}(O) = (1 + \beta_{int}(\cos(O + \omega_i) + e\cos(\omega_i))\sin(i))\ Z_{sys}(O)$ (raw light shift including the line of sight Doppler at phase o)
$\beta_z(O) = \frac{\beta}{\sqrt{1-e^2}}(\cos(O + \omega_i) + e\cos(\omega_i))\sin(i)$ (line of sight light shift)
$i =$ (orbital inclination in relation to line of sight and orbital plane)
$\omega_i =$ (phase turn or argument of periapsis)
$K_i = \frac{\beta}{\sqrt{1-e^2}}\sin(i) = \beta_{int}\sin(i)$ (semi-amplitude invariant)
$Z_{rawmax} = Z_{sys}(-\omega_i)(1 + K_i(1 + e\cos\omega_i))$
$Z_{rawmin} = Z_{sys}(\pi - \omega_i)(1 + K_i(-1 + e\cos\omega_i))$
$B_{ai} = (B_a + \omega_i)$
$Z_{sys}(-\omega_i) = (1 - \beta^2\frac{1+e^2+2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2\frac{1+e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{sys}(\pi - \omega_i) = (1 - \beta^2\frac{1+e^2-2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2\frac{1-e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{raw}(-\omega_i)\tau(-\omega_i)(1 - e\cos\omega_i) + Z_{raw}(\pi - \omega_i)\tau(\pi - \omega_i)(1 + e\cos\omega_i) = 2$ (Holographic Decryption Invariant)

Background Free Parametric Equations for the Observed Orbit

$$\begin{bmatrix} x_{sky} \\ y_{sky} \\ z_{depth} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \begin{bmatrix} x_{orb} \\ y_{orb} \\ 0 \end{bmatrix}$$
$x_{sky} = \frac{\cos(O+\omega_i)}{\kappa_o^2(O)} \cdot R_s$
$y_{sky} = \frac{\sin(O+\omega_i)\cos(i)}{\kappa_o^2(O)} \cdot R_s$
$z_{depth} = \frac{\sin(O+\omega_i)\sin(i)}{\kappa_o^2(O)} \cdot R_s$

Un-tilted 2D coordinates within the orbital plane

$x_{orb} = \frac{\cos(O+\omega_i)}{\kappa_o^2(O)} \cdot R_s$
$y_{orb} = \frac{\sin(O+\omega_i)}{\kappa_o^2(O)} \cdot R_s$

2.1 Translation to Legacy Tensor Parameters

The continuous multi-component tensor fields of General Relativity are derived as localized matrix representations of the purely scalar relational projections (β, κ) on the S^1 and S^2 carriers. The pseudo-Riemannian manifold is not a primitive; it is the coordinate-inflated shadow of relational phase capacity.

1. Metric Tensor ($g_{\mu\nu}$) The 4D metric tensor maps directly to the operational phase capacities and derived coordinate ratios.

- **Temporal Component:** $g_{tt} \mathrel{\hat{=}} -\kappa_X^2 c^2 = -c^2(1-\kappa^2)$
- **Radial Spatial Component:** $g_{rr} \mathrel{\hat{=}} \kappa_X^{-2} = (1-\kappa^2)^{-1}$
- **Transverse Spatial Components:** $g_{\theta\theta} \mathrel{\hat{=}} r^2$ and $g_{\phi\phi} \mathrel{\hat{=}} r^2 \sin^2\theta$ (where $r = R_s/\kappa^2$)

2. Four-Velocity (U^μ) and Momentum (P^μ) The 4-vector velocity translates to the composite of the local energy scale (τ) and the separated kinematic amplitudes (β_R, β_T).

- **Time Component:** $U^t = \frac{dt}{d\tau} \mathrel{\hat{=}} \frac{1}{\tau} = \frac{1}{\beta_Y \kappa_X}$
- **Spatial Components:** $U^r \mathrel{\hat{=}} c\beta_R U^t$ and $U^\phi \mathrel{\hat{=}} \frac{c}{r}\beta_T U^t$
- **Four-Momentum Norm:** $P_\mu P^\mu = -(mc)^2 \mathrel{\hat{=}} -E_0^2(\beta_Y^{-2} - \beta^2\beta_Y^{-2}) = -E_0^2$ (Strict geometric closure of the S^1 carrier).

3. Stress-Energy Tensor ($T_{\mu\nu}$) The classical sources of the gravitational field map directly to the relational density saturation limit and the intrinsic surface tension derived.

- **Energy Density:** $T_0^0 = \rho c^2 \mathrel{\hat{=}} \rho_{\max}\kappa^2 c^2 = \frac{\kappa^2 c^4}{8\pi G r^2}$
- **Isotropic Pressure:** $T_i^i = P \mathrel{\hat{=}} -\rho c^2 = -\frac{\kappa^2 c^4}{8\pi G r^2}$ (The relational tension required for $d\kappa^2/dr$ consistency as derived in Part I Density, Mass, and Pressure section).

4. Ricci Tensor ($R_{\mu\nu}$), Ricci Scalar (R), and Einstein Tensor ($G_{\mu\nu}$) The differential curvature operators are bypassed entirely by the Unified Geometric Field Equation linking geometric scale directly to energy density ratios.

- **Vacuum ($R_{\mu\nu} = 0$):** $\mathrel{\hat{=}} \frac{d}{dr}(r\kappa^2) = 0 \implies \kappa^2 = \frac{R_s}{r}$
- **Matter ($G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$):** $\mathrel{\hat{=}} \frac{d}{dr}(r\kappa^2) = \frac{8\pi G}{c^2}r^2\rho_{matter}$

5. Christoffel Symbols ($\Gamma^\lambda_{\mu\nu}$) and Geodesic Integration The affine connections ($\Gamma^\lambda_{\mu\nu}$) used to define parallel transport on a curved manifold, the continuous integration of the geodesic equation ($\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0$) are bypassed in favor of the algebraic Energy-Symmetry Law ($\Delta E_{A\to B} + \Delta E_{B\to A} = 0$) and the relational phase divergence $\Delta\varphi = 2\pi\frac{\tau_Y^2}{1-e^2}$.

2.2 Comparative Structural Analysis: Newtonian Mechanics, General Relativity, and R.O.M.

Feature	Newtonian Mechanics	General Relativity	Relational Orbital Mechanics
Primitives	Euclidean space $\mathbb{R}^3$, time t, mass M, gravitational force.	4D pseudo-Riemannian manifold $(\mathcal{M}, g_{\mu\nu})$, metric, affine connection.	Projections $\beta \in S^1$, $\kappa \in S^2$; no background manifold.
Dynamics	$\ddot{\mathbf{r}} = -\frac{GM}{r^3}\mathbf{r}$.	$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$.	$\kappa^2 = 2\beta^2$ and $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$.
Formalism	Vector calculus, potential theory.	Tensor calculus, non-linear PDEs ($G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$).	Dimensionless algebra on S^1 and S^2.
Precession $\Delta\phi$	0 (Laplace-Runge-Lenz conservation).	$\approx \frac{6\pi GM}{c^2 a(1-e^2)}$ (1PN expansion).	$\frac{2\pi(3\beta^2-2\beta^4)}{1-e^2} = \frac{3\pi R_s}{a(1-e^2)} - \frac{\pi R_s^2}{a^2(1-e^2)}$.
Core Boundary	Divergence as $r \to 0$ ($F \to \infty$).	Curvature singularity at $r = 0$; horizon at $r = R_s$.	Cutoff at $r_{\min} = R_s/2$ via $\beta^2_{\max} = 1 \implies \kappa^2_{\max} = 2$; $r = 0$ is excluded.
Dimensional Scales	G, M enter dynamics directly.	G, M couple stress-energy to curvature.	G, M are scale factors (S_X); unitless core is autonomous.

1. Algebraic Closure vs. Differential Integration

- Newtonian mechanics and GR define orbits by integrating differential equations over coordinate time.
- GR requires solving the geodesic equation via variational action $\delta\int ds = 0$ along curved connections.

- R.O.M. replaces differential evolution with algebraic closure: trajectories are equilibrium states fixed by $\kappa^2 = 2\beta^2$ and projection balances on S^1 and S^2.
- All orbital parameters evaluate directly without numerical integration.

2. Secular Precession and Higher-Order Residuals

- Leading-order post-Newtonian perihelion advance:

$$\Delta\phi_{\text{GR, 1PN}} = \frac{3\pi R_s}{a(1-e^2)}$$

- Second-order post-Newtonian (2PN) geodesic expansions introduce positive, eccentricity-dependent terms:

$$\Delta\phi_{\text{GR, 2PN}} = \frac{3\pi R_s}{a(1-e^2)} + \frac{3\pi R_s^2}{4a^2(1-e^2)^2}(18-e^2) + \mathcal{O}(R_s^3)$$

- R.O.M. derives precession directly from normalized phase divergence $\tau_Y^2 = 1-(1-\kappa^2)(1-\beta^2) = 3\beta^2 - 2\beta^4$:

$$\Delta\phi_{\text{ROM}} = \frac{2\pi\tau_Y^2}{1-e^2} = \frac{3\pi R_s}{a(1-e^2)} - \frac{\pi R_s^2}{a^2(1-e^2)}$$

- The leading term reproduces GR 1PN.
- The second-order term introduces a negative correction, $-\frac{\pi R_s^2}{a^2(1-e^2)}$, generated by carrier cross-coupling $\beta^2\kappa^2$.
- For Mercury, this correction contributes -7.306×10^{-7} arcsec/century.
- For S2 around Sgr A*, the difference enters at order $\beta^4 \approx 1.6 \times 10^{-9}$, four orders of magnitude below observational sensitivity.

3. State-Space Boundaries vs. Singularities

- GR contains geodesic incompleteness at $r = 0$, where curvature invariants diverge.
- R.O.M. bounds admissible states through kinematic saturation on S^1: $\beta^2_{\text{max}} = 1$.
- Applying closure $\kappa^2 = 2\beta^2$ fixes the minimum physical radius:

$$\kappa^2_{\text{max}} = 2 \implies r_{\text{min}} = \frac{R_s}{\kappa^2_{\text{max}}} = \frac{R_s}{2}$$

- The coordinate origin $r = 0$ is outside the relational domain.
- At the horizon ($\kappa^2 = 1$), the relational phase factor $\kappa_X = \sqrt{1-\kappa^2} \to 0$ nullifies causal transfer ($\tau = \kappa_X\beta_Y \to 0$) without infinite densities.

4. Relational Factorization

- Standard orbital mechanics embeds dimensional constants (G, c) and mass (M) directly into equations of motion.
- R.O.M. factorizes all observables into a unitless core and a scale monomial:

$$X = \hat{X}(\beta, \kappa, e, O, i, \omega_i, \kappa_R) \cdot S_X(R_s; c, G)$$

- Kinematics, geometry, and relativistic corrections are evaluated within the dimensionless core $\hat{X}$.
- Constants G and M serve solely as conversion factors to historical unit systems.

3 Relational Origin of:

In this section we derive all major orbital elements, invariants and identities. This continue our derivation chain started from 4 Foundational Methodological Principles and revealing relational origin of orbital mechanics.

Ontological Independence and Generative Derivation

Primary Goal: Predict empirically observed physical reality exclusively from the relational necessities that remain after unproven absolute structures are removed.

Descriptive vs. Generative Ontology:

- **Classical mechanics is descriptive:** it postulates absolute background containers, mass parameters (M, m), empirical constants (G), and force laws to map observations.
- **WILL Relational Geometry is generative:** physical laws emerge naturally as unavoidable consequences of pure ontology.

Structural Mechanism:
Orbital mechanics is generated solely by the phase exchange between the 1-DOF directional carrier (S^1) and the 2-DOF omnidirectional carrier (S^2), under the singular constraint of Relational Closure ($\kappa^2 = 2\beta^2$).

The standard descriptive equations are revealed as collapsed artifacts of this primary relational topology.

3.1 Eccentricity

Eccentricity is a measure of the projectional deviation from the circular equilibrium state ($\delta = 1$).

Theorem 3.1 (Geometric Eccentricity). *For a closed orbital system governed by the projection invariants of WILL Relational Geometry, the orbital eccentricity e is strictly determined by the closure factor at periapsis, δ_p:*

$$e = \frac{2\beta_p^2}{\kappa_p^2} - 1 = \frac{1}{\delta_p} - 1. \tag{2}$$

Proof. Instead of relying on classical force laws, we derive this relation directly from the conservation of the two fundamental projection invariants of the WILL framework:

1. **Kinetic Projection Invariant (Binding Energy):** $\beta^2 = \kappa_o^2(o) - \beta_o^2(o) = \text{const}$.
2. **Angular Projection Invariant:** $h = r_o(o)\beta_T(o)c = \text{const}$ ($\beta = \beta_T(o)$ at turning points).

Consider the two turning points of a closed orbit: periapsis (p) and apoapsis (a). By operational definition of the shape parameter e, the relation between radii is determined by the geometric range:

$$r_a = r_p(\frac{1+e}{1-e}). \tag{3}$$

Step 1: Relational Mapping. Using the angular invariant h (implying $\beta \propto 1/r$) and the field definition $\kappa^2 \propto 1/r$, we express the apoapsis projections in terms of the periapsis values:

$$\beta_a^2 = [\beta_p(\frac{r_p}{r_a})]^2 = \beta_p^2(\frac{1-e}{1+e})^2, \tag{4}$$

$$\kappa_a^2 = \kappa_p^2(\frac{r_p}{r_a}) = \kappa_p^2(\frac{1-e}{1+e}). \tag{5}$$

Note: Kinematic projection scales quadratically with the radius ratio, while potential projection scales linearly.

Step 2: Energy Balance. Substituting these into the kinetic global invariant β^2:

$$(\kappa_p^2 - \beta_p^2) = (\kappa_a^2 - \beta_a^2).$$

Substituting the mappings from Step 1:

$$\kappa_p^2 - \beta_p^2 = \kappa_p^2(\frac{1-e}{1+e}) - \beta_p^2(\frac{1-e}{1+e})^2.$$

Rearranging to group potential terms (κ) on the left and kinematic terms (β) on the right:

$$\kappa_p^2[1 - \frac{1-e}{1+e}] = \beta_p^2[1 - (\frac{1-e}{1+e})^2].$$

Step 3: Algebraic Reduction. Expanding the terms in brackets:

$$\text{LHS bracket } (\kappa \text{ term}): \quad 1 - \frac{1-e}{1+e} = \frac{(1+e)-(1-e)}{1+e} = \frac{2e}{1+e}.$$
$$\text{RHS bracket } (\beta \text{ term}): \quad 1 - \frac{(1-e)^2}{(1+e)^2} = \frac{(1+e)^2-(1-e)^2}{(1+e)^2} = \frac{4e}{(1+e)^2}.$$

Substituting back into the balance equation:

$$\kappa_p^2\left(\frac{2e}{1+e}\right) = \beta_p^2\left(\frac{4e}{(1+e)^2}\right).$$

Dividing both sides by $2e$ and multiplying by $(1+e)^2$:

$$\kappa_p^2(1+e) = 2\beta_p^2.$$

This yields geometric identity for bound orbits:

$$2\beta_p^2 = \kappa_p^2(1+e). \tag{6}$$

Step 4: Connection to Closure. Recall the definition of the closure factor at periapsis:

$$\delta_p = \frac{\kappa_p^2}{2\beta_p^2}.$$

Substituting Eq.(3.1) into this definition:

$$\delta_p = \frac{\kappa_p^2}{\kappa_p^2(1+e)} = \frac{1}{1+e}.$$

Solving for e, we obtain the stated result:

$$\boxed{e = \frac{1}{\delta_p} - 1 = \frac{2\beta_p^2}{\kappa_p^2} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2}}$$

□

Remark 3.2. *This result confirms that eccentricity is strictly a measure of the energetic deviation from the circular equilibrium state ($\delta = 1$), derived entirely from the conservation of relational projections without invoking mass or Newtonian forces.*

SUMMARY

$$\frac{2\beta_p^2}{\kappa_p^2} - 1 \equiv e \equiv 1 - \frac{2\beta_a^2}{\kappa_a^2}$$

ECCENTRICITY ≡ CLOSURE DEFECT

SPACETIME ≡ ENERGY

3.2 Vis-Viva

In elliptical systems, the global closure theorem $\kappa^2 = 2\beta^2$ is conserved across the entire orbital cycle, while local phase variations manifest as an internal "breathing" of relational projections. We now derive the conservation law governing this local dynamic distribution.

Proposition 3.3 (Global Kinetic Amplitude)**.** *The difference between the square of the local potential projection $\kappa_o^2(o)$ and the square of the local kinetic projection $\beta_o^2(o)$ is a global binding invariant kinetic amplitude equal to β^2.*

Proof. The binding parameter β^2 defines the global relational depth of the system:

$$\kappa^2 = 2\beta^2$$

From the definitions of the S^2 and S^1 projections at any arbitrary phase O:

$$\kappa_o^2(o) = \frac{2\beta^2(1+e\cos o)}{1-e^2}$$

$$\beta_o^2(o) = \frac{\beta^2(1+e^2+2e\cos o)}{1-e^2}$$

Subtracting the local kinetic projection from the local potential projection:

$$\kappa_o^2(o) - \beta_o^2(o) = \frac{2\beta^2 + 2\beta^2 e \cos o - (\beta^2 + \beta^2 e^2 + 2\beta^2 e \cos o)}{1 - e^2}$$

$$\kappa_o^2(o) - \beta_o^2(o) = \frac{\beta^2(1 - e^2)}{1 - e^2} = \beta^2$$

Thus, the algebraic distance between the omnidirectional carrier S^2 and directional carrier S^1 relations remains globally constant irrespective of orbital phase and shape. □

Theorem 3.4 (The Orthogonal Signature of the Orbit)**.** *The square of the local potential projection $\kappa_o^2(o)$ on the S^2 carrier is the Pythagorean square sum of the local radial $\beta_R^2(o)$, transverse $\beta_T^2(o)$ and the global β^2 kinetic projections.*

Proof. The total local kinetic projection on the S^1 carrier splits orthogonally into radial and transverse components within the orbital plane:

$$\beta_o^2(o) = \beta_R^2(o) + \beta_T^2(o)$$

Substituting this orthogonal decomposition into the phase-invariant relation $\kappa_o^2(o) - \beta_o^2(o) = \beta^2$:

$$\kappa_o^2(o) - (\beta_R^2(o) + \beta_T^2(o)) = \beta^2$$

Rearranging yields the three-dimensional algebraic relational closure:

$$\boxed{\kappa_o^2(o) = \beta_R^2(o) + \beta_T^2(o) + \beta^2}$$

□

Remark 3.5 (Ontological Replacement of Vis-Viva)**.** *This geometric invariant replaces the descriptive Newtonian Vis-Viva equation ($v^2/2 - GM/r = -GM/2a$). Standard mechanics interprets this balance as a scalar subtraction of an abstract potential energy from kinetic energy. In WILL, it is revealed as a strict Pythagorean theorem of relational geometry: the local potential projection (κ_o^2) is generated by the orthogonal summation of the kinetic "breathing" ($\beta_R^2 + \beta_T^2$) and the global S^1 projection (β^2). SPACETIME ≡ ENERGY.*

3.3 Classical Acceleration

By the conservation of the global kinetic amplitude (Vis-Viva), the spatial gradients of the potential and kinetic projections balance to maintain the relational closure of the system. Taking the spatial derivative of the orbital invariant $\kappa_o^2 - \beta_o^2 = \beta^2$ with respect to the radial distance r yields:

$$\frac{d}{dr}(\kappa_o^2) = \frac{d}{dr}(\beta_o^2) \tag{7}$$

Substituting the definitions $\kappa_o^2 = \frac{R_s}{r}$ and $\beta_o^2 = \frac{v^2}{c^2}$ into the gradient balance:

$$\frac{d}{dr}\left(\frac{R_s}{r}\right) = \frac{1}{c^2}\frac{d(v^2)}{dr} \tag{8}$$

Evaluating the derivatives provides the geometric relationship between the potential scale and the kinematic spatial variation:

$$-\frac{R_s}{r^2} = \frac{2v}{c^2}\frac{dv}{dr} \tag{9}$$

Classical mechanics introduces the external time differential dt to evaluate this spatial variation. Applying the chain rule $a_{acc} = \frac{dv}{dt} = v\frac{dv}{dr}$ maps the kinematic spatial gradient to the classical acceleration (a_{acc}):

$$-\frac{R_s}{r^2} = \frac{2}{c^2}a_{acc} \tag{10}$$

Isolating a_{acc} yields the classical Newtonian formula:

$$a_{acc} = -\frac{R_s c^2}{2r^2} = -\frac{GM}{r^2} \tag{11}$$

Within the system, energy conservation is required by the closure theorem (sec:closure) as an invariant resource of state change projected on kinetic S^1 and potential S^2 carriers. Thus, there is no need to introduce any new primitives such as force or spacetime curvature. Geometric conservation of relational energy states provides a complete operational and ontological explanation of observed phenomena.

3.4 Angular Momentum Conservation

The conservation of specific angular momentum h is derived as a phase-independent structural invariant of the relational balance.

Proposition 3.6 (Invariant Ratio of Projections)**.** *The transverse kinetic projection $\beta_T(o)$ is strictly proportional to the local potential projection $\kappa_o^2(o)$ scaled by the global system constants.*

Proof. From the definition of the local potential projection (based on closure theorem $\kappa^2 = 2\beta^2$ and geometry of ellipse):

$$\kappa_o^2(o) = 2\beta^2 \frac{(1 + e\cos o)}{1 - e^2} \implies (1 + e\cos o) = \kappa_o^2(o)\frac{1 - e^2}{2\beta^2}$$

The transverse kinetic projection $\beta_T(o)$ is defined on the S^1 carrier as:

$$\beta_T(o) = \beta \frac{1 + e\cos o}{\sqrt{1 - e^2}}$$

Substituting the expression for $(1 + e\cos o)$ into the definition of $\beta_T(o)$:

$$\beta_T(o) = \beta(\kappa_o^2(o)\frac{1 - e^2}{2\beta^2})\frac{1}{\sqrt{1 - e^2}} = \kappa_o^2(o)\frac{\sqrt{1 - e^2}}{2\beta}$$

Rearranging to isolate the invariant ratio of the phase-dependent projections:

$$\frac{\beta_T(o)}{\kappa_o^2(o)} = \frac{\sqrt{1 - e^2}}{2\beta}$$

□

Theorem 3.7 (Conservation of Angular Momentum)**.** *The specific angular momentum h, defined as the product of the local scale $r_o(o)$ and the transverse kinetic projection, is a global phase invariant.*

Proof. Using the local scale $r_o(o) = R_s/\kappa_o^2(o)$, we define h:

$$h = r_o(o)\beta_T(o)c = \frac{R_s}{\kappa_o^2(o)}\beta_T(o)c$$

Substituting the invariant ratio $\frac{\beta_T(o)}{\kappa_o^2(o)} = \frac{\sqrt{1-e^2}}{2\beta}$:

$$\boxed{h = R_s c \frac{\sqrt{1 - e^2}}{2\beta}}$$

Since R_s, e, and β are global invariants, h does not depend on phase o. Angular momentum is thus the manifestation of the fixed structural exchange rate between the S^2 and the S^1 carriers and their projections. □

3.5 Kepler's Third Law

The Keplerian proportionality $a \propto T^{2/3}$ is a direct algebraic consequence of the energetic equilibrium between the 1-DOF directional kinematic carrier (S^1) and the 2-DOF omnidirectional potential carrier (S^2). The derivation strictly utilizes dimensionless ratios and absolute geometric scale, eliminating dependence on classical mass and the empirical gravitational constant.

The kinematic amplitude β on S^1 defines (sec:SR interval) the ratio of directional motion to the universal limit c:

$$\beta = \frac{2\pi a}{cT} \tag{12}$$

The omnidirectional potential amplitude squared κ^2 on S^2 is defined (sec:GR interval) by the absolute spatial geometric scale invariant R_s and the radial distance a:

$$\kappa^2 = \frac{R_s}{a} \tag{13}$$

Energetic closure dictates the invariant geometric exchange rate between the carriers:

$$\kappa^2 = 2\beta^2 \tag{14}$$

Substituting the parameterizations into the closure condition yields:

$$\frac{R_s}{a} = 2(\frac{2\pi a}{cT})^2 = \frac{8\pi^2 a^2}{c^2 T^2} \tag{15}$$

Rearranging to isolate the semi-major axis a gives the closed-form relation:

$$a^3 = \frac{R_s c^2}{8\pi^2} T^2 \tag{16}$$

$$a = (\frac{R_s c^2}{8\pi^2})^{1/3} T^{2/3} \implies a \propto T^{2/3} \tag{17}$$

3.6 Orbital Precession as Phase Accumulation

In standard General Relativity, orbital precession is derived via perturbative methods and Taylor expansions of the Schwarzschild metric. In WILL Relational Geometry, precession is derived strictly from the algebraic accumulation of the internal phase difference over a closed geometric cycle, without differential approximations.

Theorem 3.8 (Precession Law)**.** *The secular angular shift (precession) $\Delta\varphi$ per closed orbit is the strict geometric projection of the system's relational phase difference τ_Y normalized by the orbital shape factor $(1-e^2)$.*

Proof. **Step 1: State Difference.** A system completely at rest with the observer possesses maximal internal phase $\tau = 1$. The true relational phase of an orbiting system is the product of the kinetic S^1 and potential S^2 projections carriers:

$$\tau^2 = \beta_Y^2 \kappa_X^2 = (1-\beta^2)(1-\kappa^2). \tag{18}$$

Expanding this product algebraically yields:

$$\tau^2 = 1 - (\beta^2 + \kappa^2) + \kappa^2\beta^2. \tag{19}$$

The total phase divergence from the rest origin ($\tau = 1$) is defined as:

$$\tau_Y^2 \equiv 1 - \tau^2 = (\beta^2 + \kappa^2) - \kappa^2\beta^2. \tag{20}$$

The cross-coupling term $\kappa^2\beta^2$ accounts for the non-linear geometric interaction between the directional S^1 and omnidirectional S^2 relational carriers.

Step 2: Geometric Accumulation of Phase Divergence. The system accumulates this state mismatch over the full cycle (2π). The total angular shift is the base phase scaled by the state divergence, normalized by the orbital shape factor $(1-e^2)$.

Restricted by the Mathematical Transparency principle (pr:mathematical), we do not truncate or simplify this connection. The orbital precession is locked in its , fully non-linear algebraic form:

$$\boxed{\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e^2} = \frac{2\pi(1-\tau^2)}{1-e^2}} \tag{21}$$

□

Remark 3.9 (Epistemic Purity)**.** *This derivation contrasts with the complex integration of perturbed differential equations, revealing the algebraic evaluation of the relational phase. The cross-coupling term $\kappa^2\beta^2$ distinguishes RG from a mere approximation and structurally completes the invariant topology (full mathematical proof: 8).*

3.7 Dynamic Event Horizon

In classical literature, the event horizon is often heuristically treated as an escape-velocity boundary where spacetime curvature becomes infinitely steep. In RG, the horizon emerges strictly as the limit of geometric precessional invariant, representing the ultimate topological closure of a trajectory.

Using the relational phase accumulation law (derived in 3.6), the baseline precession per orbit is given by:

$$\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e^2} = \frac{2\pi(1-\tau^2)}{1-e^2} \tag{22}$$

For a massive system under total relational closure ($\kappa^2 = 2\beta^2$), consider the **Dynamic Horizon** state where the potential projection reaches absolute saturation ($\kappa^2 = 1$). By the closure rule, the kinematic projection is $\beta^2 = \frac{1}{2}$.

At this limit, the gravitational phase component collapses ($\kappa_X = \sqrt{1-1} = 0$), causing the entire internal causal structure of the system to vanish:

$$\tau^2 = (1-\beta^2)(1-\kappa^2) = 0 \quad \Longrightarrow \quad \tau_Y = 1 \tag{23}$$

For a perfectly circular orbit ($e = 0$) at this causal limit, the precession evaluates to:

$$\Delta\varphi = \frac{2\pi \cdot (1)}{1-0} = 2\pi \tag{24}$$

Event Horizon

An orbital precession of $\Delta\varphi = 2\pi$ radians per revolution implies that the periapsis shifts by a full circle during a single orbit. Consequently, the orbital progression continuously folds onto itself perfectly, creating a closed, self-intersecting locus where forward orbital motion mathematically nullifies itself. This provides a novel interpretation of the event horizon phenomena not as a "point of infinite density" in an external container-like spacetime, but the unitary topological closure of the relational trajectory itself.

3.8 Gravitational Deflection and Lensing

In General Relativity, the deflection of light is obtained by integrating the null geodesic equations over a curved spacetime manifold. Within WILL Relational Geometry (RG), we employ neither a background manifold nor series approximations.

The system consists exclusively of its participants: the Source, the Lens (at periapsis p), and the Receiver. The total deflection angle must be derived as algebraic difference between their measurable relational phase states. The distances D_{LS} (Lens-Source), D_L (Receiver-Lens), and D_S (Receiver-Source) are defined strictly as flat relational baselines measured by the Receiver.

Unified Interaction Gradient and Spatial Scaling

Energy projection distribution among axes of relational carriers must obey the conservation law. There must exist a strict, algebraically closed gradient connecting all states. This requires a fundamental geometric symmetry between the kinematic phase buffer (governed by β) and the spatial potential phase buffer (governed by κ_X).

Theorem 3.10 (Symmetric Phase Buffer Gradient)**.** *The geometric scaling factor Γ dictates the distribution of the projections onto the spatial trajectory. Just as the kinematic projection β_p depletes the internal temporal phase, the potential projection κ_p depletes the internal spatial phase. The geometric gradients are strictly symmetric:*

$$\Gamma(\beta_p) = \frac{1+\beta_p^2}{2} \quad \text{and} \quad \Gamma(\kappa_{Xp}) = \frac{1+\kappa_{Xp}^2}{2} = 1 - \frac{\kappa_p^2}{2} \tag{25}$$

Using this symmetric gradient, we define the **Unified Closure Defect** for any trajectory as the ratio of local projections weighted by their respective phase buffers:

$$\delta_\varphi = \left(\frac{\kappa_p}{\beta_p}\right)^2 \frac{\Gamma(\beta_p)}{\Gamma(\kappa_{Xp})} = \frac{\kappa_p^2(1+\beta_p^2)}{\beta_p^2(2-\kappa_p^2)} \tag{26}$$

The geometric shape parameter (eccentricity) of the trajectory is derived directly from this defect:

$$e_\varphi = \frac{1}{\delta_\varphi} - 1 = \frac{\beta_p^2(2-\kappa_p^2)}{\kappa_p^2(1+\beta_p^2)} - 1 \tag{27}$$

Applying the algebraic transit equation for a distant observer, we arrive at the unified equation for gravitational deflection:

$$\boxed{\Delta\varphi = 2\arcsin\left(\frac{\kappa_p^2(1+\beta_p^2)}{\beta_p^2(2-\kappa_p^2) - \kappa_p^2(1+\beta_p^2)}\right)} \tag{28}$$

Verification of Topological Limits:

- **Newtonian Limit ($\beta_p \ll 1, \kappa_p \ll 1$):** The spatial buffer is full ($\Gamma(\kappa_{Xp}) \to 1$). The eccentricity is dominated by $2\beta_p^2/\kappa_p^2$. The deflection reduces to classical Rutherford scattering: $\Delta\varphi \approx \frac{\kappa_p^2}{\beta_p^2}$.
- **Photonic Limit ($\beta_p = 1$):** The kinematic phase buffer is completely exhausted ($\Gamma(\beta_p) = 1$). Substituting $\beta_p = 1$ yields the light eccentricity:
$$e_\gamma = \frac{2-\kappa_p^2}{2\kappa_p^2} - 1 = \frac{2-3\kappa_p^2}{2\kappa_p^2} \tag{29}$$
- **The Photon Sphere ($e_\gamma = 0$):** Setting $e_\gamma = 0$ identifies the topological limit where light is trapped in a perfectly circular orbit.
$$2 - 3\kappa_p^2 = 0 \quad \Longrightarrow \quad \kappa_p^2 = \frac{2}{3} \tag{30}$$
This yields the physical radius $r_p = 1.5R_s$, matching the General Relativistic photon sphere natively without a metric.

Relational Baselines and the Capture Shadow

For a distant Receiver, the relational distance to the trajectory at the moment of flyby (the optical impact parameter b) must be scaled by the complement potential phase κ_{Xp} to account for the local spatial depletion:

$$b = \frac{r_p}{\kappa_{Xp}} = \frac{R_s}{\kappa_p^2\sqrt{1-\kappa_p^2}} \tag{31}$$

At the photon sphere limit ($\kappa_p^2 = 2/3$), this scaling yields the capture shadow radius:

$$b = \frac{1.5R_s}{\sqrt{1-2/3}} = 1.5\sqrt{3}R_s \approx 2.598R_s \tag{32}$$

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Gravitational Lensing and the Einstein Ring

In Relational Geometry, the description of gravitational lensing arises entirely as the geometric projection of the deflection angle onto the Receiver's baseline plane. The potential projection κ_p^2 is determined by the physical impact parameter b, which is related to the observed angle θ_I via the Receiver-Lens baseline: $b = D_L\theta_I$.

Theorem 3.11 (Algebraic Einstein Ring)**.** *For a perfectly aligned system ($\theta_S = 0$), the image forms a symmetric ring. The angular radius of this Einstein ring, θ_E, valid for all velocities and field strengths, is given by the implicit algebraic equation:*

$$\theta_E = 2\frac{D_{LS}}{D_S}\arcsin\left(\frac{\kappa_p^2(\theta_E)(1+\beta_p^2)}{\beta_p^2(2-\kappa_p^2(\theta_E)) - \kappa_p^2(\theta_E)(1+\beta_p^2)}\right) \tag{33}$$

where $\kappa_p^2(\theta_E)$ is algebraically locked to the observable baselines via $(D_L\theta_E)^2 = \frac{R_s^2}{\kappa_p^4(1-\kappa_p^2)}$.

Remark 3.12. *Testing showed that this lensing and deflection formalisation are only approximations. The derivation of exact solution is an ongoing research.*

3.9 Schwarzschild Metric

The Schwarzschild metric is the coordinate-inflated projection of the potential phase κ_{Xo} and kinematic amplitudes β_R, β_T scaled for coordinate reference. The fundamental primitive is the dimensionless algebraic phase factor of the relational state $\tau_o^2 = \kappa_{Xo}^2\beta_{Yo}^2$. The metric is recovered by appending substantival posits.

(i) **A spatial container.** A background manifold is imposed to host the geometry.

(ii) **Externally flowing time.** An autonomous coordinate time t is introduced.

(iii) **Coordinate grid.** Kinematic projections are decomposed onto imposed spatial axes r and ϕ, transforming dimensionless amplitudes into differential coordinate ratios.

(iv) **Arbitrary scale reference.** The scalar composition is multiplied by an external magnitude c^2dt^2.

By Conservation of Relation Theorem the relational projections map to the posited coordinate differentials through the local potential phase scaling:

- **Temporal potential phase:** $\kappa_{Xo}^2 = 1 - \kappa_o^2 = 1 - \frac{R_s}{r}$
- **Radial coordinate velocity:** $\frac{dr}{cdt} = \kappa_{Xo}^2\beta_R$ (an extra $\kappa_X o$ is dictated by changing potential with radial motion)
- **Transverse coordinate velocity:** $\frac{rd\phi}{cdt} = \kappa_{Xo}\beta_T$

Substituting these coordinate functions into the standard Schwarzschild metric interval evaluates directly to the pure relational phase factorization:

$$(\frac{d\tau}{dt})^2 = (1 - \frac{R_s}{r}) - \frac{(\frac{dr}{cdt})^2}{1-\frac{R_s}{r}} - (\frac{rd\phi}{cdt})^2 \tag{34}$$

$$\tau_o^2 = \kappa_{Xo}^2 - \frac{(\kappa_{Xo}^2\beta_R)^2}{\kappa_{Xo}^2} - (\kappa_{Xo}\beta_T)^2 \tag{35}$$

$$\tau_o^2 = \kappa_{Xo}^2 - \kappa_{Xo}^2\beta_R^2 - \kappa_{Xo}^2\beta_T^2 = \kappa_{Xo}^2(1-\beta_R^2-\beta_T^2) \tag{36}$$

$$\boxed{\tau_o^2 = \kappa_{Xo}^2\beta_{Yo}^2} \tag{37}$$

Applying the arbitrary scale reference c^2dt^2 recovers the classical line element format:

$$-ds^2 = c^2(1-\frac{R_s}{r})dt^2 - (1-\frac{R_s}{r})^{-1}dr^2 - r^2d\phi^2 \tag{38}$$

The irreducible primitive

The 4D pseudo-Riemannian metric tensor $g_{\mu\nu}$ carries more ontological baggage than the relations it encodes. The irreducible primitive remains the geometric composition of the relational phase projections on the S^1 and S^2 carriers.

Schwarzschild metric:

$$\underbrace{\tau_o^2 = \kappa_{Xo}^2\beta_{Yo}^2}_{\text{irreducible primitive}} \quad + \quad \underbrace{x, y, z, t...}_{\text{coordinate inflation}} \quad = \underbrace{(\frac{d\tau}{dt})^2 = (1-\frac{R_s}{r}) - \frac{(\frac{dr}{cdt})^2}{1-\frac{R_s}{r}} - (\frac{rd\phi}{cdt})^2}_{\text{mathematical scaffolding}}.$$

Colab Sympy Verification

Schwarzschild metric sympy.ipynb

4 Intrinsic Unitless Core: Relational Parameterization

The preceding derivations in Relational Orbital Mechanics (R.O.M.) have demonstrated that the entire system can be factorized into a dimensionless core and a single scale parameter. This section formalizes this unitless formulation, proving that physical laws operate independently of anthropocentric units, metrics, or dimensional conventions and constants, and subsequently validates these principles empirically through the precise derivation of solar system orbits and precession.

4.1 The Intrinsic Relational Core: Unitless Formulation

The claim is structural: the closed R.O.M. system factorizes into a *fully solved dimensionless core* and a *one-parameter scale map*. Units, and the constants c and G that service them, belong exclusively to the map. A civilisation with different rulers, clocks, calendars and mathematical conventions solves the identical core and obtains the identical pure numbers; only the final labelling differs. Physical laws operate before — and independently of — anthropocentric units, metrics and scales.

Theorem 4.1 (Factorization of R.O.M.). *Every quantity X of the closed R.O.M. system (1) factorizes as*

$$\boxed{X \;=\; \hat{X}(\beta,\,\kappa,\,e,\,O,\,i,\omega_i,\,\kappa_R)\;\cdot\;S_X(R_s;\,c,\,G)}$$

where $\hat{X}$ is a closed-form pure number determined entirely by dimensionless relational inputs, and S_X is a monomial scale factor depending only on the quantity class:

Quantity class	S_X	***Quantity class***	S_X
projections, shifts, angles, phases	1	*frequency*	c/R_s
length	R_s	*specific angular momentum*	$R_s\,c$
time	R_s/c	*mass*	$R_s \cdot c^2/G$
velocity	c	*density*	$1/R_s \cdot c^2/G$

No dimensional constant enters any $\hat{X}$.

Corollary 4.2 (Universal Geometric Mass). *In its own geometric units every bound system has the same mass, $\hat{M} = \frac{1}{2}$. Mass is not a property of the relational state at all: it is the scale itself, $M = \frac{1}{2}\,c^2 R_s/G$ — the anchor R_s dressed in the kilogram convention. This sharpens the Absolute Scale Derivation section argument (6).* ***Epistemic Mandate: the "mass parameter" carries zero relational information beyond the system scale.***

Remark 4.3. *In this section we use (β, e) as the intrinsic input pair, simply as an example. The system is fully algebraically closed and any parameters can be used as an input.*

The normalization also exposes the Universal Horizon Constant (5) as the pure number $\hat{\tau}_{D(limit)} = 2\sqrt{2}\,\pi$ at $\kappa_R \to 1$.

4.2 Observational Input Channels

The example intrinsic pair (β, e) is reachable through algebraically independent menus of direct observables. All entries are pure ratios — frequency ratios, angle ratios, angular-speed ratios, cycle-count ratios — identical for any observer regardless of local units ("cross-cultural invariants").

e-channel (orbit shape) — any one:	
E1 apparent angular speeds	$e = \dfrac{\sqrt{\omega_{max}/\omega_{min}} - 1}{\sqrt{\omega_{max}/\omega_{min}} + 1}$
E2 apsidal Doppler pair	$e = \dfrac{\beta_p - \beta_a}{\beta_p + \beta_a}$
E3 apsidal sky-angle ratio	$e = \dfrac{r_a/r_p - 1}{r_a/r_p + 1}$
E4 balance-point phase	$e = -\cos(B_a)$
E5 projections only	$e = \frac{2\beta_p^2}{\kappa_p^2} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2}$
β-channel (orbit depth) — any one:	
B1 transverse Doppler at a	$\beta^2 = 1 - (1 + z_\beta)^{-2}$
B2 gravitational redshift at a	$\beta^2 = \frac{1}{2}(1 - (1 + z_\kappa)^{-2})$
B3 combined shift $Z_{sys} = 1/\tau$	$\beta^2 = \frac{1}{4}(3 - \sqrt{1 + 8 \cdot Z_{sys}(B_{a,d})^{-2}})$
B4 apsidal Doppler pair	$\beta = \sqrt{\beta_p \beta_a}$
B5 two-point (vis-viva)	$\beta^2 = \dfrac{(\beta_1^2 - \beta_2^2)\,(r_2/r_1)}{r_2/r_1 - 1} - \beta_1^2$
B6 surface channel	$\beta^2 = \frac{1}{2}(1 - (1 + z_{surf})^{-2}) \sin\theta_{obs}$

Channels B3 and E4 exploit the balance points: at $B_a = \arccos(-e)$ the measured light signal $\tau(B_a)$ equals the global τ, so a single spectroscopic reading there yields $\kappa^2 = \frac{1}{2}(3 - \sqrt{1 + 8\tau^2})$ (6.2Balance Point Theorem) while the phase location itself yields e.

4.3 Operational Algorithm

Stage 0 Measure pure ratios. Select one e-channel and one β-channel observable set from the menus above (plus κ_R via z_{surf} if surface outputs are wanted). No unit is recorded.

Stage 1 Invert to the intrinsic state (β, e).

Stage 2 Solve the global block: $\kappa, \kappa_X, \beta_Y, \tau, \tau_Y, Q, \hat{a}, \hat{r}_p, \hat{r}_a, \kappa_p, \beta_p, \kappa_a, \beta_a, \delta_p, \delta_a, e_X, e_Y, \Delta\phi, \Omega, \hat{T}, \hat{\omega}, \hat{h}, B_a, B_d, \hat{\rho}, \hat{M} = \frac{1}{2}$ — all closed-form pure numbers.

Stage 3 Phase-resolve at any O: $\eta_o, \hat{r}_o, \kappa_o, \beta_o, \beta_R, \beta_T, \delta_o, Q_o, \hat{\omega}_o, \tau_o, Z_{sys}(O), \zeta(O), \hat{\Delta}_{to}$.

Stage 4 Observer-project with (i, ω_i): K_i, β_z, Z_{raw}, sky-plane parametric orbit — still pure numbers (sky lengths in units of $\hat{a}$).

Stage 5 (Optional) Inject one anchor. A single dimensional reading — chronometric T giving $R_s = Tc\beta^3/\pi$, or astrometric r_p giving $R_s = r_p\kappa_p^2$ — scales the entire solved system: lengths $\times R_s$, times $\times R_s/c$, frequencies $\times c/R_s$, mass $= \frac{1}{2}c^2 R_s/G$. The system was already solved at Stage 2; this stage only names the answers in local units.

Corollary 4.4 (Cross-Cultural Invariance)**.** *Two civilisations with incommensurable rulers and clocks conventions (the concept of mass isn't mandatory), observing the same bound system through any of the channels above, compute identical intrinsic states, identical $\hat{X}$ for every quantity, and identical dimensionless predictions ($\Delta\phi$ per orbit, Z_{sys}, e, phase timings ζ). They disagree only on Stage 5 labels, and their labels are related by unit conversion. The physical content of R.O.M. is therefore invariant not merely under coordinate changes and scales, but under the choice of metrology itself.*

Test it yourself

COLAB: ROM_UNITLESS_TEST.ipynb — 50-digit verification: intrinsic identity battery, dimensional-vs-unitless commutation across 41 orders of magnitude in R_s, unit-system invariance including randomly generated alien units, c/G scrambling, ten observational routes converging on one state, and the Mercury/Jupiter parameterization executed unitlessly with the anchor injected last.
DESMOS: https://www.desmos.com/calculator/gr7qykzicf interactive user-friendly R.O.M. solver.

4.4 Relational Parameterization: Empirical Validation

Theorem 4.5 (Relational Parameterization)**.** *The complete structural and dynamical parameterization of any closed orbital system - specifically its geometric shape (e - eccentricity), secular phase shift ($\Delta\varphi$ - orbital precession), horizon scale (R_s - Schwarzschild radius), and global scale (a - semi-major axis) - are strictly determined by the algebraic relations of its dimensionless spectroscopic and chronometric observables. The derivation requires no independent mass parameter (M), gravitational constant (G), or a priori spatial metric, differential or tensor formalisms.*

Direct algebraic derivation via spectroscopic and chronometric ratios.

Proof. The proof is constructive and algebraically closed. It generalizes to any bound orbital system. We demonstrate this using the Mercury-Sun system as an operational test case and Jupiter-Sun as confirmation test. All governing equations are derived exclusively from the internal topology of the WILL relational carriers (S^1, S^2), strictly bypassing Newtonian mechanics, Special Relativity, or General Relativity. The derivation is executed using only four dimensionless, epistemologically direct observables ($e, \theta, T_{ratio}, z_{sun}$):

INPUTS (Cross-Cultural Invariants)

The following inputs are strict dimensionless ratios. They represent direct physical observables that remain identical for any observer in the Universe, regardless of local unit systems (meters/seconds) or theoretical paradigms.

Parameter	**Value**	**Direct Observational Method**
Sun's physical observables		
$\theta_\odot$	0.0046524 rad pure ratio	**Visual Size:** The angular radius of the central body in the sky. [?]
z_{sun}	2.1224×10^{-6} pure ratio	**Spectroscopic Tension:** The raw fractional frequency shift of light $(1-\nu_{obs}/\nu_{emit})$. Defined strictly as an optical geometric observable, independent of mass or Newtonian constants [?].
Mercury's physical observables		
$T_{\text{ratioM}} = T_M/T_\oplus$	0.2408 pure ratio	**Cycle Ratio:** The observed system's full orbital period divided by the observer's local orbital period. [?]
e_M	0.2056 pure ratio	**Visual Kinematics:** Derived directly from the ratio of the planet's fastest (ω_{max}) and slowest (ω_{min}) apparent angular speeds: $e = \frac{\sqrt{\omega_{max}/\omega_{min}}-1}{\sqrt{\omega_{max}/\omega_{min}}+1}$ [?].
Jupiter's physical observables		
$T_{\text{ratio}J} = T_J/T_\oplus$	11.862 pure ratio	**Cycle Ratio:** The full Jovian orbital period measured against the observer's local (Earth) orbital period — a purely temporal, clock-based observable requiring no unit conversion or gravitational constant [?].
e_J	0.04839266 pure ratio	**Visual Kinematics:** Derived directly from the ratio of Jupiter's fastest (ω_{max}) and slowest (ω_{min}) apparent angular speeds along its orbit: $e = \frac{\sqrt{\omega_{max}/\omega_{min}}-1}{\sqrt{\omega_{max}/\omega_{min}}+1}$ [?].

Algebraic Derivation

1. Relational Scale Factor

The ratio of the Sun's radius to the planet's perihelion distance is derived entirely from visual size and cycle ratios:

$$R_{ratio} = \frac{\theta_\odot}{(T_M/T_\oplus)^{2/3} \cdot (1 - e_M)} \approx 0.0151290$$

2. Potential Projection (at perihelion)

Translating the spectral shift at the Sun's surface (z_{sun}) to the potential projection (**??**) at Mercury's perihelion (κ_p^2) using the derived scale factor:

$$\kappa_p^2 = (1 - \frac{1}{(1 + z_{sun})^2}) \cdot R_{ratio} \approx 6.4219 \times 10^{-8}$$

3. Kinematic Projection (at perihelion)

Deriving orbital kinematics strictly from the potential and eccentricity (3.1), requiring no Doppler or radar data:

$$\beta_p^2 = \frac{\kappa_p^2}{2}(1 + e_M) \approx 3.8712 \times 10^{-8}$$

4. Global Orbital Invariants

Extracting the system's global kinetic projection (β) via the invariant binding energy $\beta^2 = \kappa_p^2 - \beta_p^2$ (via Energy-Symmetry Law sec:energy-symmetry):

$$\beta^2 \approx 2.5508 \times 10^{-8} \quad \text{and} \quad \kappa^2 = 2\beta^2 \approx 5.1016 \times 10^{-8}$$

(via Closure theorem thm:closure).

5. Orbital Precession

The secular phase shift emerges (3.6) from the accumulation of the relational state divergence ($\tau_Y^2 = 1 - (1-\kappa^2)(1-\beta^2)$) over a full cycle:

$$\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1 - e_M^2} \approx 5.0203 \times 10^{-7} \text{ rad/orbit}$$

Converting this pure geometric phase shift to standard observational units yields **43.015 arcsec/century**, closely matching the observed precession of Mercury.

6. Absolute System Scale (Metric Translation)

To translate the geometric scale into legacy metric units (meters), we map Mercury's orbital period ($T_M = 7,600,521.6$ s) and the speed of light (c) to the global kinematic projection (β):

$$R_{sRG} = \frac{\beta^3}{\pi} \cdot T_M c \approx 2954.8 \text{ m}$$

Standard GR value:

$$R_{sGR} = \frac{2GM_{sun}}{c^2} \approx 2953.33 \text{ m} \quad \text{where} \quad M_{sun} = 1.98847 \times 10^{30} \text{ kg}$$

Discrepancy:

$$\frac{R_{sGR} - R_{sRG}}{R_{sGR}} \times 100 = |0.0476|\%$$

This variance is strictly within the observational uncertainty limits of the input parameters (z_{sun} and $\theta_\odot$).

7. Semi-major Axis (Standard value $\approx 5.79 \times 10^{10}$ m):

$$a = \frac{R_{sRG}}{\kappa^2} \approx 5.792288 \times 10^{10} \text{ m}$$

8. Perihelion Radius (Standard value $\approx 4.60 \times 10^{10}$ m):

$$r_p = \frac{R_{sRG}}{\kappa_p^2} \approx 4.601394 \times 10^{10} \text{ m}$$

Universality of the Framework: The Jovian Extension

To rigorously demonstrate that this parameterization is not a localized artifact of the Mercury-Sun proximity, but a universal geometric property of bound systems, we apply the exact same algebraic chain to Jupiter. We replace the inputs with Jupiter to Earth orbital periods ratios (T_{ratioJ}) and Jupiter's eccentricity (e_J), keeping the central reference invariants ($\theta_\odot$, z_{sun}, $T_\oplus$) strictly identical.

Jovian Inputs:

- $T_J = 4332.589$ days [?]
- $e_J = 0.04839266$ [?]
- $T_{ratioJ} = T_J/T_\oplus \approx 11.862$ [?]

1. Global Potential Projection (κ_J)

The local potential projection at Jupiter's semi-major axis reduces to a remarkably elegant relation, scaling the solar surface tension strictly through the inverse temporal cycle ratio:

$$\kappa_J = \sqrt{(1 - \frac{1}{(1+z_{sun})^2}) \cdot \frac{\theta_\odot}{(T_{ratioJ})^{2/3}}} \approx 6.1623 \times 10^{-5}$$

2. Absolute System Scale Recovery (R_{sRGJ})

Expanding the algebraic chain utilizing Jupiter's specific kinematic projection (β_J) yields the invariant core scale:

$$R_{sRGJ} = \frac{T_J \cdot c}{\pi} [\frac{1}{1-e_J}(1 - \frac{1}{(1+z_{sun})^2})\frac{\theta_\odot}{T_{ratioJ}^{2/3}} - \frac{1}{2}\frac{1+e_J}{1-e_J}(1 - \frac{1}{(1+z_{sun})^2})\frac{\theta_\odot}{T_{ratioJ}^{2/3}}]^{3/2}$$

$$R_{sRGJ} \approx 2955.41 \text{ m}$$

Jovian Orbital Recovery

Translating the recovered systemic scale to the spatial scale of Jupiter's orbit:

$$a_J = \frac{R_{sRGJ}}{\kappa_J^2} \approx 7.7827 \times 10^{11} \text{ m}$$

Observed value $a_{JDATA} = 778.57 \times 10^9$ m.

$$\text{Discrepancy} = \frac{|a_{JDATA} - a_J|}{a_{JDATA}} \times 100 \approx 0.038\%$$

The reconstruction of the Schwarzschild radius (R_s) and the orbital scale (a) from disparate chronometric cycles (T, e) demonstrates that there is a constant, direct, invariant relation between:

- **potential** geometry (κ^2 - **"spacetime structure"**)
- and system's **kinetic** (β^2- **"energy dynamics"**) both functioning as relational measures of state difference, derived directly from the foundational unification principle **SPACETIME ≡ ENERGY** (pr:unifying).

□

Test it yourself

DESMOS: https://www.desmos.com/calculator/mjen4ms452
COLAB: Mercury-Sun_Relational_Parameterization.ipynb

Conclusion

All major parameters of gravitational systems are recovered through algebraic closure without mass (M), gravitational constant (G), metric manifold, differential formalism, or external temporal flow.

Corollary 4.6 (Epistemic Mandate and Parameter Parsimony)**.** *If a system's generative chain is algebraically complete without a specific parameter, that parameter is an emergent scaling constant rather than a fundamental primitive. Because the structural and dynamical states $(e, \Delta\varphi, R_s, a)$ are closed using only dimensionless spectroscopic and chronometric ratios $(e, \theta_\odot, T_{ratio}, z_{sun})$, the variables G and M function as translation factors for legacy unit systems. Their retention is required exclusively for mapping invariant relational geometry onto the kilogram-based convention without increasing the system's predictive capacity.*

COLAB: M and G independence full test

ROM_MASS_G_FREE_CLOSURE.ipynb

5 Time-Density Invariant (τ_D)

Definition 5.1 (Time-Density Invariant)**.** *The time-density invariant τ_D establishes the systemic phase-space limit for any stable trajectory, defined through the orbital period T and visual angular radius θ:*

$$\tau_D = T \sin(\theta)^{\frac{3}{2}} \tag{39}$$

Theorem 5.2 (Relational Density)**.** *The structural density of the central body is determined by the time-density invariant τ_D, bypassing the Newtonian spatial mass distribution.*

Proof. For any stable trajectory, geometry enforces the trigonometric identity $\sin(\theta) = \frac{R_{sur}}{a}$. The operational parameter τ_D algebraically defines the theoretical orbital cycle at the physical surface R_{sur} of the central body:

$$\tau_D = \frac{2\pi R_{sur}}{\beta_{sur} c} \tag{40}$$

where β_{sur} is the local kinematic projection at the surface. Relational density ρ as derived in WILL RG Part I : operates via spatial projection limits:

$$\rho = \frac{\kappa^2 c^2}{8\pi G a^2} \tag{41}$$

Isolating the invariant τ_D structures this relationship as the inverse determinant of the system's central density:

$$\tau_D = \sqrt{\frac{\pi}{G\rho}} \tag{42}$$

□

This shows that information about the density of the central body is encoded in the invariant product of directly observable period and angular radius for any stable orbit in the system:

$$\tau_D = T\sin(\theta)^{\frac{3}{2}} \tag{43}$$

Theorem 5.3 (Universal Horizon Constant)**.** *At the event horizon boundary limit, the proportional ratio of the limiting time-density $\tau_{D(limit)}$ to the system scale R_s yields a universal horizon constant, operating exclusively on the universal tempo c and geometric constant π.*

Proof. The systemic horizon scale R_s isolates mathematically from the time-density invariant and surface kinematic projection β_{sur} :

$$R_s = \frac{c}{\pi}\tau_D\beta_{sur}^3 \tag{44}$$

At the event horizon boundary limit ($r = R_s$), the potential projection reaches maximum ($\kappa^2 = 1$) and the kinematic projection fixes via the closure theorem ($\beta_{sur}^2 = \frac{1}{2}$). The time-density limit evaluates to:

$$\tau_{D(limit)} = 2\pi\sqrt{2}\frac{R_s}{c} \tag{45}$$

The proportional ratio resolves to the universal geometric constant length ratio:

$$\boxed{\frac{\tau_{D(limit)}c}{R_s} = 2\pi\sqrt{2}} \tag{46}$$

□

Remark 5.4 (Ontological Consequence)**.** *The algebraic isolation of $\frac{\tau_{D(limit)}c}{R_s} = 2\pi\sqrt{2}$ demonstrates that the boundary mechanics of WILL operate exclusively on geometric proportionality and kinematic cycles. The parameters G and M are entirely absent from the limiting relational structure of the system.*

6 Algebraic Determination of the Horizon Scale

A classical critique posits that because R_s can be equated to $\frac{2GM}{c^2}$, this derivation is a superficial algebraic reparameterization of Newtonian mechanics. As shown in 9, 8, sec:earth-gps and ROM_FULL_TEST.ipynb, - this objection is contradicted by the mathematical results: it reverses the causal and ontological hierarchy.

The terms G and M are epistemologically secondary composites. They function as anthropic translation factors required to map Relational Geometry onto an assumed absolute spacetime container. R_s, conversely, is a primary, coordinate-independent relational invariant determent by the tension of the system's potential difference.

The derivation isolates true dynamic proportionality using exclusively operational observables (chronometric phase T) and intrinsic horizon scaling (a, R_s). Relational Geometry does not inherit from Newtonian constructs; rather, classical mechanics collapses from Relational Geometry when an absolute spatial background is imposed. R_s does not conceal mass; rather, classical mass is an anthropocentric proxy for the invariant relational horizon R_s.

6.1 Method A: Differential (Two-Point Method)

This method is suitable when the orbital period is unknown, but the trajectory can be traced geometrically.

Theorem 6.1 (Two-Point Schwarzschild Scale)**.** *Given measurements of geometric position (r) and kinematic intensity (β) at two arbitrary points along a trajectory:*

- *Radii: r_1, r_2 (from astrometry)*
- *Intensities: β_1, β_2 (from de-projected spectral line widths or proper motion)*

the Schwarzschild radius is:

$$\boxed{R_s = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2).} \tag{47}$$

Proof. We invoke the **Conservation of the Global Kinetic Amplitude** (Vis-Viva Theorem), which states that for any closed system, the difference between the local potential and local kinetic projections is a global invariant equal to β^2.

Therefore, at any two points 1 and 2:

$$\kappa_1^2 - \beta_1^2 = \kappa_2^2 - \beta_2^2. \tag{48}$$

Rearranging to group the projections:

$$\kappa_1^2 - \kappa_2^2 = \beta_1^2 - \beta_2^2. \tag{49}$$

Substituting the field identity $\kappa^2 = R_s/r$:

$$\frac{R_s}{r_1} - \frac{R_s}{r_2} = \beta_1^2 - \beta_2^2. \tag{50}$$

Factoring out the scale parameter R_s:

$$R_s(\frac{r_2 - r_1}{r_1 r_2}) = \beta_1^2 - \beta_2^2. \tag{51}$$

Solving for R_s:

$$R_s = \frac{\beta_1^2 - \beta_2^2}{\frac{r_2 - r_1}{r_1 r_2}} = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2). \tag{52}$$

This formula extracts the system scale purely from geometric gradients. □

6.2 Method B: Geometric Resonance (Balance Point Method)

Using one of two orbital balance points:

- Ascending: $B_a = \arccos(-e)$,
- Descending $B_d = 2\pi - \arccos(-e)$,

when the system passes through its geometric balance points the instantaneous radius r equals the semi-major axis a. At these unique phases, the closure condition $\kappa_o^2 = 2\beta_o^2$ is satisfied.

Theorem 6.2 (Balance Point Formula)**.** *Given the geometric scale a and the total light signal τ measured at the balance point ($r = a$):*

$$\boxed{R_s = \frac{a}{2}(3 - \sqrt{1 + 8\tau^2(B_a)}).} \tag{53}$$

$$B_a = \arccos(1 - \delta^{-1}) = \arccos(-e) = \arccos(1 - \tfrac{2\beta_p^2}{\kappa_p^2}) = \arccos(\tfrac{2\beta_a^2}{\kappa_a^2} - 1) \text{ (orbital balance point where } \kappa_o^2 = 2\beta_o^2 \text{ is true)}$$

Proof. **Step 1: Spacetime Factor Expansion.** At the balance point ($r = a$), the closure condition implies $\kappa^2 = R_s/a$ and $\beta^2 = R_s/2a$. The observable τ is:

$$\tau^2 = (1 - \kappa^2)(1 - \beta^2) = (1 - \frac{R_s}{a})(1 - \frac{R_s}{2a}). \tag{54}$$

Step 2: Exact Geometric Solution. Expanding the product:

$$\tau^2 = 1 - \frac{R_s}{2a} - \frac{R_s}{a} + \frac{R_s^2}{2a^2} = 1 - \frac{3R_s}{2a} + \frac{R_s^2}{2a^2}. \tag{55}$$

Multiplying by $2a^2$ to clear denominators and rearranging into standard quadratic form ($Ax^2 + Bx + C = 0$):

$$R_s^2 - 3aR_s + 2a^2(1 - \tau^2) = 0. \tag{56}$$

Solving for R_s using the quadratic formula ($x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a_{coef}}$):

$$R_s = \frac{3a \pm \sqrt{(3a)^2 - 4(1)(2a^2(1 - \tau^2))}}{2}. \tag{57}$$

Simplifying the term under the radical:

$$9a^2 - 8a^2(1 - \tau^2) = 9a^2 - 8a^2 + 8a^2\tau^2 = a^2(1 + 8\tau^2). \tag{58}$$

Thus:

$$R_s = \frac{3a \pm a\sqrt{1 + 8\tau^2}}{2} = \frac{a}{2}(3 \pm \sqrt{1 + 8\tau^2}). \tag{59}$$

For stable orbits, we must select the negative root. □

Remark 6.3 (Light Separation)**.** *This unit-less factor in brackets is:*

- ***Potential projection at semi-major axis:*** $\kappa^2 = \frac{1}{2}(3 - \sqrt{1 + 8\tau^2(B_a)})$

A unique way to separate gravitational part from the light signal.

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6.3 Method C: Instantaneous (Arbitrary Phase Method)

Most generally, if the orbital geometry (a) is known, the scale R_s can be derived from a **single epoch** observation at any arbitrary radius r_o.

Theorem 6.4 (Arbitrary Phase Formula)**.** *Given the orbital geometry (a, r_o) and the single light observable τ_o:*

$$\boxed{R_s = \frac{r_o}{2(2a - r_o)}(4a - r_o - \sqrt{(4a - r_o)^2 - 8a(2a - r_o)(1 - \tau_o^2)}).} \tag{60}$$

Proof. **Step 1: Relational Invariant Form.** We start with the global kinetic amplitude invariant $\beta^2_{global} = \kappa_o^2 - \beta_o^2$. For a bound orbit, the global kinetic amplitude is $\beta^2_{global} = R_s/2a$. We express the local kinematic intensity β_o^2 at arbitrary radius r in terms of the local field intensity κ_o^2:

$$\kappa_o^2 - \beta_o^2 = \frac{R_s}{2a} \implies \beta_o^2 = \kappa_o^2 - \frac{R_s}{2a}. \tag{61}$$

Substituting $\kappa_o^2 = R_s/r$:

$$\beta_o^2 = R_s(\frac{1}{r} - \frac{1}{2a}). \tag{62}$$

Step 2: Constraint via Observables. We substitute the expressions for κ_o^2 and β_o^2 into the observable constraint $\tau_o^2 = (1 - \kappa_o^2)(1 - \beta_o^2)$:

$$\tau_o^2 = (1 - \frac{R_s}{r})(1 - R_s[\frac{1}{r} - \frac{1}{2a}]). \tag{63}$$

Let $A = \frac{1}{r}$ and $B = \frac{1}{r} - \frac{1}{2a}$. The equation becomes:

$$\tau_o^2 = (1 - AR_s)(1 - BR_s) = 1 - (A + B)R_s + ABR_s^2. \tag{64}$$

Rearranging into quadratic form:

$$(AB)R_s^2 - (A + B)R_s + (1 - \tau_o^2) = 0. \tag{65}$$

Step 3: Coefficient Expansion. We compute the coefficients explicitly:

$$AB = \frac{1}{r}(\frac{2a - r}{2ar}) = \frac{2a - r}{2ar^2} \tag{66}$$

$$A + B = \frac{1}{r} + \frac{2a - r}{2ar} = \frac{2a + 2a - r}{2ar} = \frac{4a - r}{2ar} \tag{67}$$

Multiplying the entire quadratic equation by $2ar^2$ to clear denominators:

$$(2a - r)R_s^2 - r(4a - r)R_s + 2ar^2(1 - \tau_o^2) = 0. \tag{68}$$

Step 4: Solution. Solving for R_s:

$$R_s = \frac{r(4a - r) \pm \sqrt{r^2(4a - r)^2 - 4(2a - r)(2ar^2)(1 - \tau_o^2)}}{2(2a - r)}. \tag{69}$$

Factoring out r^2 from the radical term $\sqrt{r^2(...)} = r\sqrt{(...)}$:

$$R_s = \frac{r}{2(2a - r)}((4a - r) \pm \sqrt{(4a - r)^2 - 8a(2a - r)(1 - \tau_o^2)}). \tag{70}$$

For stable orbits, we select the negative root, yielding the algebraic link between the observed light phase and the system's geometric scale. □

7 Rotational Systems (Kerr Without Metric)

Contextual Bounds

- **For a gravitationally closed (static) system,** the physical boundary is defined by the condition $\kappa^2 = 1$. The closure principle ($\kappa^2 = 2\beta^2$) is what dictates that this corresponds to a kinetic state of $\beta^2 = 1/2$.
- **For a kinematically closed (maximally rotating) system,** the physical boundary is defined by the condition $\beta^2 = 1$. The same closure principle ($\kappa^2 = 2\beta^2$) then necessitates that the corresponding gravitational state must be $\kappa^2 = 2$.

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For rotating black holes, we establish the connection between relational kinetic projection and the Kerr metric by defining:

$$\beta = \frac{ac^2}{GM}, \quad \kappa = \sqrt{2}\beta$$

where:

- β is the relational rotation parameter, with $0 \leq \beta \leq 1$,
- κ is related to the geometry and gravity,
- $R_s = \frac{2GM}{c^2}$ is the Schwarzschild radius,
- $a = \frac{J}{Mc}$ is the Kerr rotation parameter,
- J is the angular momentum of the black hole,
- M is the mass of the black hole.

We also derive a key invariant relationship:

$$a_{\max} = \frac{R_s}{2} = \beta_{\max}^2 r$$

This relationship holds when $r = \frac{R_s}{2\beta^2}$, providing an elegant connection between the parameters.

Event Horizon

Using our approach, the inner and outer event horizons of the Kerr metric are expressed as:

$$r_{\pm} = \frac{R_s}{2}(1 \pm \beta_Y)$$

For the extreme case where $\beta = 1$ (maximal rotation), the horizons merge at:

$$r_+ = r_- = \frac{R_s}{2}$$

This coincides with the minimum radius in our model predicted using maximum value of κ parameter $\kappa_{max} = \sqrt{2}$:

$$r_{\min} = \frac{1}{\kappa_{max}^2} R_s = \frac{1}{2} R_s$$

Ergosphere

The radius of the ergosphere in our model is described as:

$$r_{\text{ergo}} = \frac{R_s}{2}(1 + \sqrt{1 - \beta^2 \cos^2\theta})$$

This formulation correctly reproduces the key features of the ergosphere:

- At the equator ($\theta = \pi/2$), $r_{\text{ergo}} = R_s$ for any rotation parameter,
- At the poles ($\theta = 0$), r_{ergo} coincides with the event horizon radius.

Ring Singularity

Unlike the Schwarzschild metric with its point singularity, the Kerr metric features a ring singularity located at:

$$r = 0, \quad \theta = \frac{\pi}{2}$$

The "size" of this ring is proportional to $a = \frac{GM}{c^2}\beta$, reaching its maximum for extreme black holes ($\beta = 1$).

Naked Singularity

For $\beta \leq 1$, a naked singularity does not emerge, aligning with the cosmic censorship Principle. In our model, the Energy Symmetry Law enforces this constraint by limiting β to the range $[0, 1]$.

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7.1 Chiral Asymmetry: The Algebraic Origin of Frame-Dragging

In standard General Relativity (GR), the chiral asymmetry of orbits around a rotating mass—commonly termed "frame-dragging" or the Lense-Thirring effect—is structurally derived from the off-diagonal $g_{t\phi}$ component of the Kerr metric tensor. In Relational Orbital Mechanics (R.O.M.), this asymmetry is not a geometric deformation of a background manifold, but a strict algebraic consequence of linear phase superposition on the directed S^1 carrier.

Composite Kinematic State and Dimensional Projection

By the Relational Origin Principle, neither the orbiting test object nor the central body possesses intrinsic velocity or spin. The only physically real quantity is the relational state difference, resolved on the 1-DOF kinematic carrier S^1.

The orbital relation constitutes a 1-DOF kinematic sequence. The rotation of a 3D central body constitutes a 3-DOF kinematic process, as change occurs simultaneously across three spatial dimensions. To evaluate the composite state, the 3-DOF spin process of central body is projected onto the 1-DOF orbit of the test object.

By the DOF-Indifference Lemma, no spatial dimension is privileged. The linear projection of a 3-DOF process onto a 1-DOF carrier requires equal distribution of amplitude across all independent degrees of freedom. The operational spin projection β_{spin} scales as $1/3$ of the specific angular momentum parameter $a = J/(Mc)$ relative to spatial depth r:

$$\beta_{\text{spin}} = \frac{a}{3r} \tag{71}$$

Because both the orbital and spin contributions share the same 1-DOF relational S^1 carrier, they superimpose linearly prior to the application of the closure theorem. The composite linear kinematic state is:

$$\beta_\Sigma = \beta_{\text{orb}} \pm \beta_{\text{spin}} \tag{72}$$

where $(+)$ denotes prograde and $(-)$ denotes retrograde alignment. Squaring is applied only to the completed S^1 amplitude β_Σ, never to its individual shares.

The Algebraic Symmetry Breaker

Applying the relational closure theorem ($\kappa^2 = 2\beta^2$) to the composite state yields:

$$\kappa^2 = 2(\beta_{\text{orb}} \pm \beta_{\text{spin}})^2 = 2\beta_{\text{orb}}^2 + 2\beta_{\text{spin}}^2 \pm 4\beta_{\text{orb}}\beta_{\text{spin}} \tag{73}$$

The emergent cross-term $\pm 4\beta_{\text{orb}}\beta_{\text{spin}}$ is the symmetry breaker. Because $\kappa^2 = R_s/r$, this mathematically mandates that a prograde orbit $(+)$ requires a deeper potential projection (smaller radius) than a retrograde orbit $(-)$ at an equivalent orbital velocity. This algebraic constraint is operationally indistinguishable from the spatial drag generated by the $g_{t\phi}$ metric.

Expansion of the Precessional Divergence

The internal causal order of the composite system evaluates via the relational phase factor $\tau^2 = (1-\kappa^2)(1-\beta_\Sigma^2)$. Expanding this algebraically and applying the strict directional boundary $(\pm = 1)$ isolates the chiral divergence:

$$\tau^2_{Y(\text{chiral})} = \pm(6\beta_{\text{orb}}\beta_{\text{spin}} - 8\beta_{\text{orb}}^3\beta_{\text{spin}} - 8\beta_{\text{orb}}\beta_{\text{spin}}^3) \tag{74}$$

Substituting the local operational projection into the isolated chiral polynomial defines the phase shift per orbital cycle, normalized by the orbital shape $(1 - e^2)$:

$$\Delta\varphi_{\text{chiral}} = \frac{2\pi}{1-e^2}[\pm(6\beta_{\text{orb}}\beta_{\text{spin}} - 8\beta_{\text{orb}}^3\beta_{\text{spin}} - 8\beta_{\text{orb}}\beta_{\text{spin}}^3)] \tag{75}$$

Colab Sympy Verification

Kerr sympy.ipynb

7.2 Legacy Translation: Lense-Thirring Equivalence

To verify structural compatibility with historical models, we isolate the primary geometric observable by ablating the higher-order geometric cross-coupling terms ($\beta^4 \to 0$) and setting the orbit to circular ($e = 0$). The equation simplifies algebraically to:

$$\Delta\varphi_{\text{ablated}} \simeq 12\pi\beta_{\text{orb}}\beta_{\text{spin}} \tag{76}$$

Substituting the operational spin projection $\beta_{\text{spin}} = \frac{J}{3Mcr}$ and the kinematic orbital projection $\beta_{\text{orb}} = \frac{v}{c}$ recovers the empirical post-Newtonian node precession:

$$\Delta\varphi_{\text{ablated}} \simeq 12\pi(\frac{v}{c})(\frac{J}{3Mcr}) = \frac{4\pi vJ}{Mc^2r} \tag{77}$$

Applying the kinematic scaling identity $\frac{v}{M} = \frac{G}{rv}$:

$$\Delta\varphi_{\text{ablated}} \simeq \frac{4\pi GJ}{c^2 r^2 v} \tag{78}$$

The Lense-Thirring effect emerges natively from the algebraic closure of the relational carriers, executed without invoking differential tensors or a curved pseudo-Riemannian manifold.

COLAB Empirical Verification

This Chiral Asymmetry when tested against LAGEOS-1 Satellite (Earth orbit) shows perfect agreement with observations and GR predictions: LAGEOS I Lense-Thirring.ipynb

7.3 Epistemological Conclusion

To generate chiral orbital asymmetry (frame-dragging), General Relativity must postulate a pseudo-Riemannian manifold with an off-diagonal metric tensor component, $g_{t\phi}$, to induce spatial torsion. R.O.M. reproduces the identical physical symmetry and precessional kinematics algebraically. Enforcing the relational closure theorem onto the composite one-dimensional carrier $\beta_\Sigma = \beta_{\text{orb}} \pm \beta_{\text{spin}}$ yields the necessary quadratic expansion. The emergent cross-term:

$$\pm 4\beta_{\text{orb}}\beta_{\text{spin}} \tag{79}$$

mathematically mandates the chiral divergence. This algebraic constraint is operationally indistinguishable from the spatial drag generated by the $g_{t\phi}$ metric. Therefore, within the relational framework, the geometric deformation of a background spacetime (as a dimensional container) and its differential tensors are ontologically and mathematically redundant.

8 Mercury's Precession: GR 1PN as First-Order Approximation of RG

8.1 Statement of the Verification Problem

The standard first-order post-Newtonian expression for the perihelion advance of a bound two-body orbit is

$$\Delta\varphi_{\text{GR}} = \frac{3\pi\, R_s}{a\,(1-e^2)}, \tag{80}$$

with a the semi-major axis, e the eccentricity, and $R_s = 2GM/c^2$ the Schwarzschild scale of the central body.

WILL Relational Geometry (WILL RG) expresses the secular angular shift per closed orbit as the strict geometric accumulation of the system's phase divergence from the rest origin:

$$\Delta\varphi_{\text{RG}} = \frac{2\pi\,\tau_Y^2}{1-e^2}, \qquad \tau_Y^2 \equiv 1-\tau^2 = 1-(1-\beta^2)(1-\kappa^2) = \beta^2+\kappa^2-\beta^2\kappa^2, \tag{81}$$

evaluated at the orbital reference scale (semi-major axis a), where the closure condition $\beta^2(a) = \kappa^2(a)/2$ and the field relation $\kappa^2(a) = R_s/a$ hold.

The question addressed in this section is whether (80) is the leading-order content of (81) in the small parameter R_s/a. The structure of (81) suggests an even stronger statement than in the time-dilation case (sec:earth-gps): the GR formula should be recoverable by removing a single explicit term — the cross-coupling $\beta^2\kappa^2$ — from τ_Y^2, with the residual being exactly $-2\pi\,\beta^2\kappa^2/(1-e^2)$.

8.2 Methodology

The verification is split into two independent parts, both executed in a single public notebook (Section 8.5).

Part I — symbolic identity. The WILL RG precession is expressed in closed form by substituting the closure condition $\beta^2 = \kappa^2/2$ and $\kappa^2 = R_s/a$ into (81). The result is simplified algebraically and subtracted from (80). The algebraic identity between the residual and the analytically predicted second-order term $-2\pi\,\beta^2\kappa^2/(1-e^2)\big|_a$ is verified using `sympy.simplify`. As an independent cross-check, (81) is Taylor-expanded in a bookkeeping parameter ε that scales R_s, and the first-order coefficient is compared with (80).

Part II — numerical comparison. Both (80) and (81) are evaluated with 40-digit precision arithmetic (`mpmath`, `mp.dps = 40`) using Mercury's J2000 orbital elements: $a = 5.7909050 \times 10^{10}\,\text{m}$, $e = 0.20563593$, orbital period $T = 87.9691\,\text{d}$, $GM_\odot = 1.32712440018 \times 10^{20}\,\text{m}^3\,\text{s}^{-2}$, $c = 299{,}792{,}458\,\text{m}\,\text{s}^{-1}$, giving $R_s^\odot = 2953.25\,\text{m}$. Per-orbit values are converted to arcseconds per Julian century using $36{,}525\,\text{d}/T$ orbits per century. The numerical discrepancy $\Delta\varphi_{\text{RG}} - \Delta\varphi_{\text{GR}}$ is compared with the analytically predicted second-order correction obtained in Part I.

8.3 Results

Symbolic. Substituting the closure and field relations into (81) yields the closed-form WILL RG precession

$$\Delta\varphi_{\rm RG} \;=\; \frac{3\pi\,R_s}{a\,(1-e^2)} \;-\; \frac{\pi\,R_s^2}{a^2\,(1-e^2)}. \tag{82}$$

The first term is identically (80). The algebraic difference is therefore

$$\Delta\varphi_{\rm RG} - \Delta\varphi_{\rm GR} \;=\; -\,\frac{\pi\,R_s^2}{a^2\,(1-e^2)} \;\equiv\; -\,\frac{2\pi\,\beta^2\kappa^2}{1-e^2}\bigg|_a. \tag{83}$$

The identity $\left[\Delta\varphi_{\rm RG}-\Delta\varphi_{\rm GR}\right]-\left[-2\pi\,\beta^2\kappa^2/(1-e^2)\right]\big|_a=0$ vanishes identically in (R_s,a,e). The Taylor expansion of (81) in R_s/a reproduces (80) at first order with zero residual.

Structural observation. In the time-dilation case the GR additive formula was the linear coefficient of a Taylor expansion of a multiplicative τ-ratio. In the present case the correspondence is more direct: the GR first-order PN precession formula coincides exactly with the WILL RG expression *after a single explicit cancellation*, namely $\beta^2\kappa^2\to 0$ in $\tau_Y^2=1-(1-\beta^2)(1-\kappa^2)$. The omitted term encodes the second-order coupling between the kinematic S^1 and gravitational S^2 projection carriers of the WILL framework.

Numerical. With Mercury's parameters listed above:

quantity	value
$\Delta\varphi_{\rm RG}$ (exact, per orbit)	$5.01867565337\times10^{-7}$ rad
$\Delta\varphi_{\rm GR}$ (first-order PN, per orbit)	$5.01867573869\times10^{-7}$ rad
$\Delta\varphi_{\rm RG}-\Delta\varphi_{\rm GR}$ (per orbit)	-8.531×10^{-15} rad
$-\pi R_s^2/[a^2(1-e^2)]$ (predicted, per orbit)	-8.531×10^{-15} rad
residual	$\sim10^{-41}$ rad
$\Delta\varphi_{\rm RG}$ (per century)	$42.9807844914''$
$\Delta\varphi_{\rm GR}$ (per century)	$42.9807852220''$
$\Delta\varphi_{\rm RG}-\Delta\varphi_{\rm GR}$ (per century)	$-7.306\times10^{-7}\,''$

The numerical discrepancy $\Delta\varphi_{\rm RG}-\Delta\varphi_{\rm GR}$ matches the analytically derived second-order term (83) to within the limit set by the 40-digit floating-point arithmetic. The relative magnitude of the term omitted by the first-order GR formula is $\sim1.7\times10^{-8}$ of the total precession.

8.4 Interpretation

Equation (82) establishes that the standard first-order post-Newtonian formula (80) is the leading-order content of the WILL RG expression (81) in R_s/a, and that the omitted contribution is exactly the cross-coupling term $-2\pi\,\beta^2\kappa^2/(1-e^2)$ at the semi-major axis. This is an algebraic identity in (R_s,a,e), not a numerical near-match.

For Mercury, the second-order term in (82) contributes at the level of $-7.3\times10^{-7}\,''\,\text{century}^{-1}$, i.e. $\sim1.7\times10^{-8}$ of the total precession of $\approx42.98''\,\text{century}^{-1}$. This deviation is well below the current precision of perihelion measurements and does not, by itself, constitute a distinguishing observation between the two formulations in the Solar System regime. The sign of the correction is fixed: WILL RG predicts a slightly smaller precession than the first-order post-Newtonian formula.

The scope of the present statement is restricted to the relation between the WILL RG closed-form expression (81) and the *first-order post-Newtonian* GR precession formula (80). A comparison against fully non-linear geodesic integration in the Schwarzschild geometry — which itself receives positive-definite corrections beyond the leading $3\pi R_s/[a(1-e^2)]$ term — is a separate calculation and is not addressed in this section.

8.5 Reproducibility

The complete symbolic and numerical verification reported in this section is contained in a single public Google Colab notebook, comprising the two scripts used to produce (82), (83), and the numerical table above:

Colab Notebook

Mercury precession RG 1st order=GR.ipynb

The notebook has no external dependencies beyond `sympy` and `mpmath` and is self-contained. All input parameters are listed explicitly in the numerical script; any change in a, e, or $GM_\odot$ rescales both $\Delta\varphi_{\rm RG}$ and $\Delta\varphi_{\rm GR}$ consistently and does not affect the symbolic identity (83).

9 S2 Star (Sgr A*): R.O.M. vs GR and Instrumental Systematics

We apply Relational Orbital Mechanics to the 24-year astrometric and radial-velocity dataset of the S2 star orbiting the Galactic Centre: 46 astrometric epochs (1995–2018) and 82 radial velocities from six instruments, 174 raw observables in total. The R.O.M. model is confronted with two General Relativity references. The first is the standard first post-Newtonian parametrisation with a time-linear pericentre advance $\omega(t) = \omega_0 + \dot{\omega}\,(t - T_0)$. The second is a direct numerical integration of the 1PN equations of motion in harmonic gauge,

$$\ddot{\vec{r}} = -\frac{GM}{r^2}[(1 + \frac{v^2}{c^2} - \frac{4GM}{c^2 r})\hat{n} \;-\; \frac{4\dot{r}\,\vec{v}}{c^2}], \tag{84}$$

which contains no parametrisation choice of any kind: the trajectory is whatever GR produces. The integration is validated independently of the fits: trajectories agree to $5 \times 10^{-4}\,\mu$as between tolerance settings and to $1.2 \times 10^{-2}\,\mu$as between integration methods, the 1PN energy returns to its value at successive pericentre passages to 1.6×10^{-12} relative, and the measured apsidal advance reproduces $6\pi GM/c^2 p$ to 1.2×10^{-4} ($12.40'$ per orbit). All three models carry the same nine free parameters $(a_{\rm as}, e, i, \omega, \Omega, P, T_0, V_0, K)$, zero free precession parameters, identical single-step Rømer correction, identical data and error model.

In R.O.M. the pericentre advance and the system scale are algebraically locked to the closure invariant:

$$f_{\rm prec} = \frac{3\beta^2 - 2\beta^4}{1 - e^2}, \qquad \beta = \frac{K\sqrt{1-e^2}}{c\,\sin i}, \qquad Z_{\rm sys} = \frac{1}{\tau_o} = \frac{1}{\sqrt{1-\kappa_o^2}\sqrt{1-\beta_o^2}}, \qquad R_s = \frac{Tc\beta^3}{\pi}, \qquad a = \frac{R_s}{2\beta^2}. \tag{85}$$

The GR 1PN advance, expressed through the same fitted observables, is $\Delta\omega_{\rm 1PN}/2\pi = 3(K/c)^2/\sin^2 i$ per orbit; at S2's velocity amplitude the two magnitudes agree to 3×10^{-5} relative, so any difference between the fits originates in trajectory geometry, not in the size of the advance.

Bare nine-parameter comparison and its resolution

Model (9 orbital parameters each)	χ^2 **bare**	χ^2 **+ frame & RV offsets**	χ^2/dof
WILL R.O.M. (closure-locked, phase-coupled)	728.6	228.3	$4.42 \to 1.46$
GR 1PN, time-linear $\omega(t)$	800.8	225.1	$4.85 \to 1.44$
GR 1PN, direct integration of motion in harmonic gauge	809.3	227.8	$4.90 \to 1.46$

Table 1: S2 fit quality. Bare fits use orbital parameters only; the right column adds the standard calibration model of published S2 analyses (astrometric zero point x_0, y_0, reference-frame drift v_{x0}, v_{y0}, five inter-instrument radial-velocity offsets), applied identically to every model and profiled exactly. dof = 165 and 156 respectively.

Taken at face value, the bare comparison favours R.O.M. by $\Delta\chi^2 = 80.7$ against integrated GR with equal parameter count. This number is arithmetically genuine and is not a property of the GR implementation: the time-linear baseline reproduces the integrated trajectory to $36\,\mu$as RMS on noise-free synthetic data, well inside the $574\,\mu$as median astrometric error, so replacing it by the integration changes the gap by only 8.5 units. The number is nevertheless not a statement about gravitation. A reduced χ^2 of 4.4–4.9 rejects all three bare models at high significance: no published analysis of this dataset fits absolute sky positions without a reference-frame solution, because the infrared frame is known to carry zero-point and drift uncertainty at the mas and $0.1\,{\rm mas\,yr^{-1}}$ level. When the standard linear calibration terms are granted identically to every model, all three converge to the same frame solution (offset ≈ 1 mas, drift 0.08–0.21 $\rm mas\,yr^{-1}$, instrument offsets 10–95 $\rm km\,s^{-1}$) and the gap closes completely: $\Delta\chi^2 = -0.5$ over 156 degrees of freedom. Relational Geometry and General Relativity are statistically indistinguishable on the S2 orbit.

Injection–recovery: mechanism, directionality, non-circularity

The origin of the bare-fit difference was established by controlled injection. Synthetic observables were generated at the real epochs with the real error model, from a known truth, and refitted by both nine-parameter models.

Injected truth	**Fit configuration**	$\Delta\chi^2$ (GR − R.O.M.)
Integrated GR, no systematics	bare, 9 par.	−12
Integrated GR + frame drift, data direction	bare, 9 par.	+50
Integrated GR + frame drift, rotated direction	bare, 9 par.	−24
R.O.M. + frame drift	with calibration model	$+18.7 \pm 11.2$
Integrated GR + frame drift	with calibration model	-5.8 ± 11.4
Real S2 data	with calibration model	**−0.5**

Table 2: Injection–recovery results (six noise realisations per calibrated row; injected drift in the data direction: $x_0 = -1.0$, $y_0 = +0.9$ mas, $v_{x0} = +0.08$, $v_{y0} = +0.20\,{\rm mas\,yr^{-1}}$). The real data fall on the GR-truth expectation and 1.7σ below the R.O.M.-truth expectation.

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Three facts follow. First, R.O.M. holds no generic advantage: on clean GR truth it loses, and it loses again when the injected drift points in a rotated direction. Second, the full observed advantage is manufactured by injecting a purely instrumental drift into purely GR data; the bare $\Delta\chi^2$ therefore measures alignment between R.O.M.'s trajectory family and this dataset's particular frame drift, not relational dynamics. Third, the calibration model is not a signal eraser and is not circular: when R.O.M. is the injected truth its advantage survives the identical procedure at $+18.7 \pm 11.2$, the frame parameters are profiled freely inside each model's own fit with no GR quantity entering the R.O.M. fit anywhere, and R.O.M.'s own fit independently demands the same frame solution, improving by 500 χ^2 units when granted it.

Rigidity, not flexibility

The absorption is a structural property of the closure, not added freedom. Fitted to noise-free integrated-GR observables, the R.O.M. family cannot approach the GR trajectory below a 440 μas RMS floor, so it is not a superset of GR and possesses no surplus fitting capacity; it carries zero free precession parameters, and the closure $\kappa^2 = 2\beta^2$ removes degrees of freedom rather than adding them. Relative to the time-linear family it owns exactly one rigid distortion mode: the apsidal advance enters as $\omega_{\text{shift}} = f_{\text{prec}}\, O$, locked to the true phase. The osculating apsis of the integrated GR orbit itself advances in the same phase-locked manner, accumulating 67% of each orbit's advance within six months of pericentre against 7% for a linear ramp, so phase-locked advance is native to GR's osculating dynamics in the Damour–Deruelle quasi-Keplerian sense; the single-eccentricity rosette is a third curve, distinct from both the time-linear approximation and the exact trajectory. That single rigid mode, projected through 24 years of epochs, happens to point along the documented frame drift of this dataset. A one-mode family absorbing a one-direction systematic by alignment is the precise mechanism by which a less flexible RG outperformed GR, and it is the reason an explicit one-line drift term cancels the effect.

What the S2 test establishes

At the precision of current S2 data, R.O.M. is observationally equivalent to General Relativity. It reaches this equivalence from a closed dimensionless algebra: the advance magnitude matches GR 1PN to 3×10^{-5}, the full physical scale follows from chronometric and spectroscopic ratios alone through $R_s = Tc\beta^3/\pi$, with G and M entering only as a posteriori unit conversions, and the complete fit is evaluated in closed form. The genuine point of departure between the frameworks is of order β^4, where $f_{\text{prec}} = (3\beta^2 - 2\beta^4)/(1-e^2)$ separates from the GR second post-Newtonian structure; at S2's $\beta^2 \approx 4 \times 10^{-5}$ this lies four orders of magnitude below the sensitivity of the dataset. The falsifiable frontier of the closure law is therefore not stellar orbits at the Galactic Centre but regimes where β^4 terms are resolved: relativistic pulsar binaries, horizon-scale orbits, and gravitational-wave inspirals.

Reproduction and MCMC Analysis

All fits, the 1PN integration, the calibration profiling, and the injection–recovery experiments are reproduced by the scripts in ROMvsGR S2 study.ipynb. Full MCMC Analysis https://willrg.com/reports/MCMC_S2.html

10 Relational Derivation of the L1 Equilibrium Point

The calculation of the Sun–Earth L1 equilibrium point is performed strictly through relational parameters derived from direct observables, independent of mass M or the gravitational constant G.

10.1 Observational Primitives

The derivation utilises only the following epistemologically direct quantities (identical in form to those used for Mercury and S2 in earlier sections):

- $\theta_\odot$: angular radius of the Sun (0.0046524 rad).
- $z_\odot$: spectroscopic fractional frequency shift at the solar surface (2.1224×10^{-6}).
- T_E: sidereal orbital period of the Earth (3.15576×10^7 s).
- T_m: sidereal orbital period of the Moon (2.3606×10^6 s).
- R_m: mean Earth–Moon distance (3.844×10^8 m).

10.2 Structural Parameters from Relational Projections

Using the absolute-scale formulae established in Section 6 (Method A) and the global kinematic relation $R_s = Tc\beta^3/\pi$:

$$R_{s\odot} = \frac{T_E}{\pi}c(\sqrt{\frac{1}{2}(1 - \frac{1}{(1+z_\odot)^2})\sin(\theta_\odot)})^3, \tag{86}$$

$$\beta_m = \frac{2\pi R_m}{T_m c}, \tag{87}$$

$$R_{s\mathrm{E}} = \frac{T_m}{\pi}c\,\beta_m^3, \tag{88}$$

$$R_E = \frac{R_{s\odot}}{(1 - \frac{1}{(1+z_\odot)^2})\sin(\theta_\odot)}. \tag{89}$$

These expressions recover the Earth–Sun distance $R_E \approx 1.496 \times 10^{11}$ m and the Earth/Sun scale ratio $\mu = R_{s\mathrm{E}}/R_{s\odot} \approx 3.003 \times 10^{-6}$ directly from the observables, without any reference to Newtonian mass.

10.3 Earth–L1 Distance Derivation

Let R_{L1} be the distance from Earth to the L1 point and define the normalised separation

$$\alpha = \frac{R_E - R_{L1}}{R_E}.$$

Synchronous motion at L1 requires the kinematic projection β_{L1} to satisfy the orbital-period constraint. Enforcing the closure theorem $\kappa^2 = 2\beta^2$ together with energy symmetry yields the quintic

$$\alpha^5 - 2\alpha^4 + \alpha^3 + \Big(\frac{R_{s\mathrm{E}}}{R_{s\odot}} - 1\Big)\alpha^2 + 2\alpha - 1 = 0.$$

In the limit $R_{s\mathrm{E}} \ll R_{s\odot}$ the physical root is

$$R_{L1} \approx R_E\sqrt[3]{\frac{R_{s\mathrm{E}}}{3R_{s\odot}}}.$$

Substituting the numerical values obtained above gives

$$R_{L1} \approx 1.498 \times 10^9\,\mathrm{m}$$

(with more precise ephemeris constants the value rises to 1.501×10^9 m). Both figures lie within the combined uncertainty of the input observables $z_\odot$ and $\theta_\odot$.

Result

The relational derivation reproduces the observed Earth–L1 distance

$$R_{L1} \approx 1.5 \times 10^9\,\mathrm{m}$$

to better than 0.3% without invoking G, M, or a background metric. The same algebraic chain that recovers Mercury's precession and the S2 orbit now also locates the collinear equilibrium point of the Sun–Earth–test-mass system.

Test it yourself

Full symbolic + numerical verification (including the quintic root): L1 Relational Derivation.ipynb

This section closes the demonstration that all major gravitational observables - two-body orbits, precession, light deflection, and now the three-body L1 point - emerge as an algebraic consequence of the Foundational Methodological Principles.

Proceed to next section of WILL Relational Geometry paper: **Density, Mass, and Pressure**